

Conflicts are parametrically encoded: initial evidence for a cognitive space view to reconcile the debate of domain-general and domain-specific cognitive control

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Abstract

Cognitive control resolves conflicts between task-relevant and -irrelevant information to enable goal-directed behavior. As conflicts can arise from different sources (e.g., sensory input, internal representations), how a limited set of cognitive control processes can effectively address diverse conflicts remains a major challenge. Based on the cognitive space theory, different conflicts can be parameterized and represented as distinct points in a (low-dimensional) cognitive space, which can then be resolved by a limited set of cognitive control processes working along the dimensions. It leads to a hypothesis that conflicts similar in their sources are also represented similarly in the cognitive space. We designed a task with five types of conflicts that could be conceptually parameterized. Both human performance and fMRI activity patterns in the right dorsolateral prefrontal (dlPFC) support that different types of conflicts are organized based on their similarity, thus suggesting cognitive space as a principle for representing conflicts.

eLife assessment

Yang et al. investigate whether distinct sources of conflict are represented in a common cognitive space. The study uses an interesting task that mixes different sources of difficulty and reports that the brain appears to represent these sources as a mixture on a continuum in prefrontal areas. While the findings could be **valuable** to theory in this area, there are concerns with the analysis, design and results, that raise uncertainty regarding the main conclusion of a shared cognitive space. Thus, the evidence reported here ranges from **solid** to **incomplete**.

Introduction

Cognitive control enables humans to behave purposefully by modulating neural processing to resolve conflicts between task-relevant and task-irrelevant information. For example, when naming the color of the word “BLUE” printed in red ink, we are likely to be distracted by the word meaning, because reading a word is highly automatic in daily life. To keep our attention on the color, we need to mobilize the cognitive control processes to resolve the conflict between the color and word by boosting/suppressing the processing of color/word meaning. As task-relevant and task-irrelevant information can come from different sources, the sources of conflicts and how they should be resolved can vary greatly (Kornblum et al., 1990 [↗](#)). For example, the conflict may occur between items of sensory information, such as between a red light and a police officer signaling cars to pass. Alternatively, conflict may occur between sensory and motor information, such as when a voice on the left asks you to turn right. The large variety of conflict sources implies that there may be innumerable number of conflict conditions. A key unsolved question in cognitive control is how our brain efficiently resolves these different types of conflicts.

A first step to addressing this question is to examine the commonalities and/or dissociations across different types of conflicts that can be categorized into different *domains*. Examples of the domains of conflicts include experimental paradigm (Freitas et al., 2007 [↗](#); Magen & Cohen, 2007 [↗](#)), sensory modality (Hazeltine et al., 2011 [↗](#); Yang et al., 2017 [↗](#)), or conflict type regarding the dimensional overlap of conflict processes (Jiang & Egner, 2014 [↗](#); Liu et al., 2004 [↗](#)).

Two solutions to resolving a wide range of conflict types are proposed. They differ based on whether the same cognitive control mechanisms are applied across domains. On the one hand, the *domain-general* cognitive control theories posit that the frontoparietal cortex adaptively encodes task information and can thus flexibly implement control strategies for different types of conflicts. This is supported by the generalizable control adjustment (i.e., encountering a conflict trial from one type can facilitate conflict resolution of another type) (Freitas et al., 2007 [↗](#); Kan et al., 2013 [↗](#)) and similar neural patterns (Peterson et al., 2002 [↗](#); Wu et al., 2020 [↗](#)) across distinct conflict tasks. A broader domain-general view holds that the frontoparietal brain regions/networks are widely involved in multiple control demands well beyond the conflict domain (Assem et al., 2020 [↗](#); Cole et al., 2013 [↗](#)), which explains the remarkable flexibility in human behaviors. However, since domain-general processes are by definition likely shared by different tasks, when we need to handle multiple task demands at the same time, the efficiency of both tasks would be impaired due to resource competition or interference (Musslick & Cohen, 2021 [↗](#)). Therefore, the domain-general processes is evolutionarily less advantageous for humans to deal with the diverse situations requiring high efficiency (Cosmides & Tooby, 1994 [↗](#)). On the other hand, the *domain-specific* theories argue that different types of conflicts are handled by distinct cognitive control processes (e.g., where and how information processing should be modulated) (Egner, 2008 [↗](#); Kim et al., 2012 [↗](#)). However, according to the domain-specific view, the highly varied conflict situations require a large variety of preexisting control processes, which is biologically implausible (Abrahamse et al., 2016 [↗](#)).

To reconcile the two theories, researchers recently proposed that cognitive control might be a mixture of domain-general and domain-specific processes. For instance, Freitas and Clark (2015) [↗](#) found that trial-by-trial adjustment of control can generalize across two conflict domains to different degrees, leading to domain-general (strong generalization) or domain-specific (weak or no generalization) conclusions depending on the task settings of the consecutive conflicts. Similarly, different brain networks may show domain-general (i.e., representing multiple conflicts) or domain-specificity (i.e., representing individual conflicts separately) (Jiang & Egner, 2014 [↗](#); Li et al., 2017 [↗](#)). Even within the same brain area (e.g., medial frontal cortex), Fu et al. (2022) [↗](#) found that the neural population activity can be factorized into orthogonal dimensions

encoding both domain-general and domain-specific conflict information, which can be selectively read out by downstream brain regions. While the mixture view provides an explanation for the contradictory findings (Braem et al., 2014), it suffers the same criticism as domain-specific cognitive control theories, as it still requires numerous cognitive control processes to fully cover all possible conflicts.

A key to reconciling domain-general and domain-specific cognitive control is to organize the large number of conflict types using a system with limited, dissociable dimensions. A construct with a similar function is the *cognitive space* (Bellmund et al., 2018), which extends the idea of cognitive map (Behrens et al., 2018) to the representation of abstract information. Critically, the cognitive space view holds that the representations of different abstract information are organized continuously and the representational geometry in the cognitive space is determined by the similarity among the represented information (Bellmund et al., 2018).

In the human brain, it has been shown that abstract (Behrens et al., 2018; Schuck et al., 2016) and social (Park et al., 2020) information can be represented in a cognitive space. For example, social hierarchies with two independent scores (e.g., popularity and competence) can be represented in a 2D cognitive space (one dimension for each score), such that each social item can be located by its score in the two dimensions (Park et al., 2020). In the field of cognitive control, recent studies have begun to conceptualize different control states within a cognitive space (Badre et al., 2021). For example, Fu et al. (2022) mapped different conflict conditions to locations in a low/high dimensional cognitive space to demonstrate the domain-general/domain-specific problems; Grahek et al. (2022) used a cognitive space model of cognitive control settings to explain behavioral changes in the speed-accuracy tradeoff. However, the cognitive spaces proposed in these studies were only applicable to a limited number of control states involved in their designs. Therefore, it remains unclear whether there is a cognitive space that can explain the large number of control states, similar to that of the spatial location (Bellmund et al., 2018) and non-spatial knowledge (Behrens et al., 2018). A challenge to answering this question lies in how to construct control states with continuous levels of similarity. Our recent work (Yang et al., 2021) showed that it is possible to manipulate continuous conflict similarity by using a mixture of two independent conflict types with varying ratios, which can be used to further examine the behavioral and neural evidence for the cognitive space view. It is also unclear how the cognitive space of cognitive control is encoded in the brain, although that of spatial locations and non-spatial abstract knowledge has been relatively well investigated in the medial temporal lobe, medial prefrontal and orbitofrontal system (Behrens et al., 2018; Bellmund et al., 2018). Recent research has suggested that the abstract task structure could be encoded and implemented by the frontoparietal network (Vaidya & Badre, 2022; Vaidya et al., 2021), but whether a similar neural system encodes the cognitive space of cognitive control remains untested.

The present study aimed to test the geometry of cognitive space in conflict representation. Specifically, we hypothesize that different types of conflicts are represented as points in a cognitive space. Importantly, the distance between the points, which reflects the geometry of the cognitive space, scales with the difference in the sources of the conflicts being represented by the points. The dimensions in the cognitive space of conflicts can be the aforementioned *domains*, in which domain-specific cognitive control processes are defined. For a specific type of conflict, its location in the cognitive space can be parameterized using a limited number of coordinates, which reflect how much control is needed for each of the domain-specific cognitive control processes. The cognitive space can also represent different types of conflicts with low dimensionality (Badre et al., 2021; MacDowell et al., 2022). Different domains can be represented conjunctively in a single cognitive space to achieve domain-general cognitive control, as conflicts from different sources can be resolved using the same set of cognitive control processes. We further hypothesize that the cognitive space representing different types of conflicts may be located in the frontoparietal network due to its essential roles in conflict resolution (Freund, Bugg, et al., 2021; Fu et al., 2022) and abstract task representation (Vaidya & Badre, 2022).

In this study, we adjusted the paradigm from our previous study (Yang et al., 2021 [DOI](#)) by including transitions of trials from five different conflict types, which enabled us to test if these conflict types are organized in a cognitive space (**Fig. 1A** [DOI](#)). Specifically, on each trial, an arrow, pointing either upwards or downwards, was presented on one of the 10 possible locations on the screen. Participants were required to respond to the pointing direction of the arrow (up or down) by pressing either the left or right key. Importantly, conflicts from two sources can occur in this task. On one hand, the vertical location of the arrow can be incongruent with the direction (e.g., an up-pointing arrow on the lower half of the screen), resulting spatial Stroop conflict (Liu et al., 2004 [DOI](#); Lu & Proctor, 1995 [DOI](#)). On the other hand, the horizontal location of the arrow can be incongruent with the response key (e.g., an arrow requiring left response presented on the right side of the screen), thus causing Simon conflict (Lu & Proctor, 1995 [DOI](#); Simon & Small, 1969 [DOI](#)). As the arrow location rotates from the horizontal axis to the vertical axis, spatial Stroop conflict increases, and Simon conflict decreases. Therefore, the 10 possible locations of the arrow give rise to five conflict types with unique blend of spatial Stroop and Simon conflicts (Yang et al., 2021 [DOI](#)). As the increase in spatial Stroop conflict is perfectly correlated with the decrease in Simon conflict, we can use a 1D cognitive space to represent all five conflict types.

One way to parameterize (i.e., defining a coordinate system) the cognitive space is to encode each conflict type by the angle of the axis connecting its two possible stimulus locations (**Fig. 1B** [DOI](#)). Within this cognitive space, the similarity between two conflict types can be quantified as the cosine value of their angular difference (**Fig. 1C** [DOI](#)). The rationale behind defining conflict similarity based on combinations of different conflict sources, such as spatial-Stroop and Simon, stems from the evidence that these sources undergo independent processing (Egner, 2008 [DOI](#); Li et al., 2014 [DOI](#); Liu et al., 2010 [DOI](#); Wang et al., 2014 [DOI](#)). Identifying these distinct sources is critical in efficiently resolving diverse conflicts. If the conflict types are organized as a cognitive space in the brain, the similarity between conflict types in the cognitive space should be reflected in both the behavior and similarity in the neural representations of conflict types. Our data from two experiments using this experimental design support both predictions: using behavioral data, we found that the influence of congruency (i.e., whether the task-relevant and task-irrelevant information indicate the same response) from the previous trial to the next trial increases with the conflict similarity between the two trials. Using fMRI data, we found that more similar conflicts showed higher multivariate pattern similarity in the right dorsolateral prefrontal cortex (dlPFC).

Results

Conflict type similarity modulated behavioral congruency sequence effect (CSE)

Experiment 1

We conducted a behavioral experiment ($n = 33$, 18 females) to examine how CSEs across different conflict types are influenced by their similarity. First, we validated the experimental design by testing the congruency effects. All five conflict types showed robust congruency effects such that the incongruent trials were slower and less accurate than the congruent trials (Note S1; Fig. S1 A/B). To test the influence of similarity between conflict types on behavior, we examined the CSE in consecutive trials. Specifically, the CSE was quantified as the interaction between previous and current trial congruency and can reflect how (in)congruency on the previous trial influences cognitive control on the current trial (Egner, 2007 [DOI](#); Schmidt & Weissman, 2014 [DOI](#)). It has been shown that the CSE diminishes if the two consecutive trials have different conflict types (Akay & Hazeltine, 2011 [DOI](#); Egner et al., 2007 [DOI](#); Torres-Quesada et al., 2013 [DOI](#)). Similarly, we tested whether the size of CSE increases as a function of conflict similarity between consecutive trials. To this end, we organized trials based on a 5 (previous trial conflict type) $\times$ 5 (current trial conflict

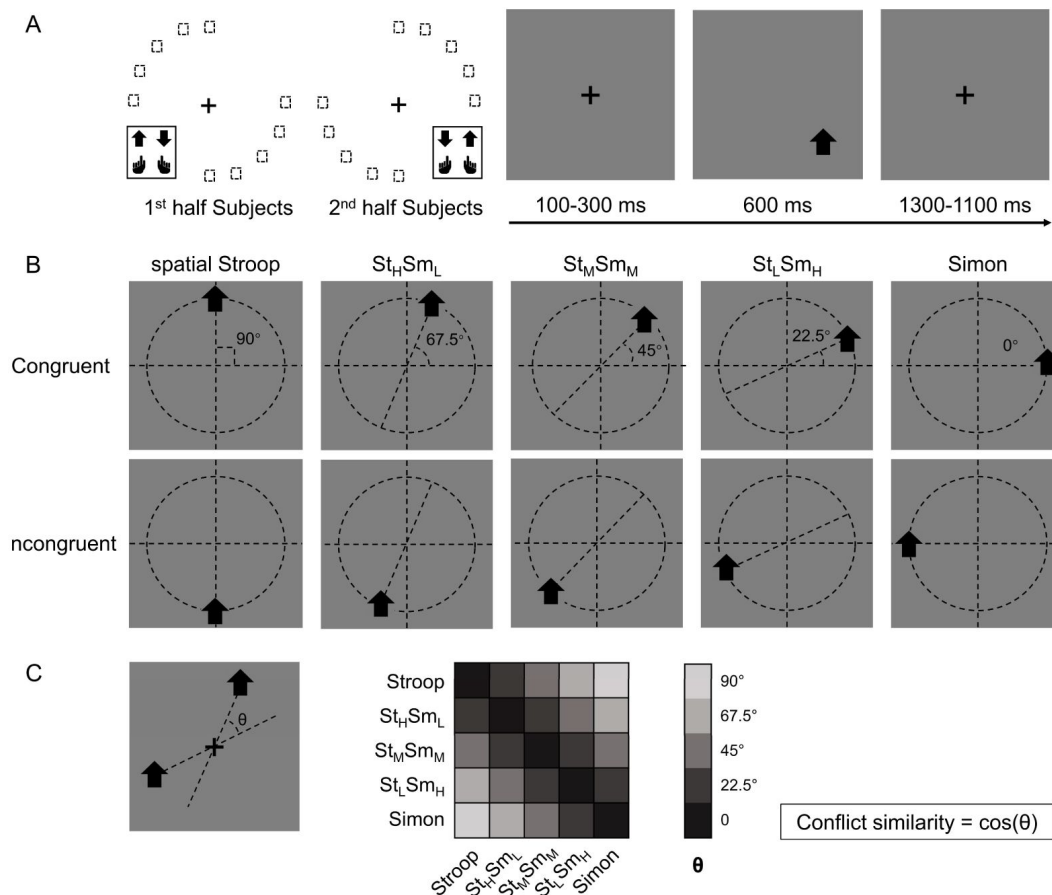


Fig. 1.

Experimental design.

(A) The left panel shows the orthogonal stimulus-response mappings of the two participant groups. In each group the stimuli were only displayed at two quadrants of the circular locations. One group were asked to respond with the left button to the upward arrow and with the right button to the downward arrow presented in the to-left and bottom-right quadrants, and the other group vice versa. The right panel shows the time course of one example trial. The stimuli were displayed for 600 ms, preceded and followed by fixation crosses that lasted for 1400 ms in total. (B) Examples of the five types of conflicts, each containing congruent and incongruent conditions. The arrows were presented at locations along five orientations with isometric polar angles, in which the vertical location introduces the spatial Stroop conflict, and the horizontal location introduces the Simon conflict. Dashed lines are shown only to indicate the location of arrows and were not shown in the experiments. (C) The definition of the angular difference between two conflict types and the conflict similarity. The angle θ is determined by the acute angle between two lines that cross the stimuli and the central fixation. Therefore, stimuli of the same conflict type form the smallest angle of 0, and stimuli between Stroop and Simon form the largest angle of 90°, and others are in between. Conflict similarity is defined by the cosine value of θ . H = high; L = low; M = medium.

type) $\times$ 2 (previous trial congruency) $\times$ 2 (current trial congruency) factorial design, with the first two and the last two factors capturing between-trial conflict similarity and the CSE, respectively. The cells in the 5×5 matrix were mapped to different similarity levels according to the angular difference between the two conflict types (**Fig. 1C**). As shown in **Fig. 2**, the CSE, measured in both reaction time (RT) and error rate (ER), scaled with conflict similarity.

To test the modulation of conflict similarity on the CSE, we constructed a linear mixed effect model to predict RT/ER in each cell of the factorial design using a predictor encoding the interaction between the CSE and conflict similarity (see Methods). The results showed a significant effect of conflict similarity (RT: $\beta = 0.10 \pm 0.01$, $t(1978) = 15.82$, $p < .001$, $\eta_p^2 = .120$; ER: $\beta = 0.15 \pm 0.02$, $t(1978) = 7.84$, $p < .001$, $\eta_p^2 = .085$, Fig. S2B/E). In other words, the CSE increased with the conflict similarity between two consecutive trials. The conflict similarity modulation effect remained significant after regressing out the influence of physical proximity between the stimuli of consecutive trials (Note S2). As a control analysis, we also compared this approach to a two-stage analysis that first calculated the CSE for each previous $\times$ current trial conflict type condition and then tested the modulation of conflict similarity on the CSEs (Yang et al., 2021). The two-stage analysis also showed a strong effect of conflict similarity (RT: $\beta = 0.58 \pm 0.04$, $t(493) = 14.74$, $p < .001$, $\eta_p^2 = .388$; ER: $\beta = 0.36 \pm 0.05$, $t(493) = 7.01$, $p < .001$, $\eta_p^2 = .320$, Fig. S2A/D). Importantly, individual modulation effects of conflict similarity were positively correlated between the two approaches (RT: $r = 0.48$; ER: $r = 0.86$, both $ps < 0.003$, one-tailed), indicating the consistency of the estimated conflict similarity effects across the two approaches. In the following texts, we will use the terms “*conflict similarity effect*” and “*conflict type effect*” interchangeably.

Moreover, to test the continuity and generalizability of the similarity modulation, we conducted a leave-one-out prediction analysis. We used the behavioral data from Experiment 1 for this test, due to its larger amount of data than Experiment 2. Specifically, we removed data from one of the five similarity levels (as illustrated by the θ s in **Fig. 1C**) and used the remaining data to perform the same mixed-effect model (i.e., the two-stage analysis). This yielded one pair of beta coefficients including the similarity regressor and the intercept for each subject, with which we predicted the CSE for the removed similarity level for each subject. We repeated this process for each similarity level once. The predicted results were highly correlated with the original data, with $r = .87$ for the RT and $r = .84$ for the ER, $ps < .001$.

Experiment 2

Behavioral results

We next conducted an fMRI experiment using a shorter version of the same task with a different sample ($n = 35$, 17 females) to seek neural evidence of how different conflict types are organized. Using behavioral data, we first validated the experimental design by testing congruency effects in each of the five conflict types (Note S1; Fig. S1 C/D). We then tested the modulation of conflict similarity on the behavioral CSE using the linear mixed effect model as in Experiment 1 (except the two-stage method). Results showed a significant effect of conflict similarity modulation (RT: $\beta = 0.24 \pm 0.04$, $t(1148) = 6.36$, $p < .001$, $\eta_p^2 = .096$; ER: $\beta = 0.33 \pm 0.06$, $t(1206) = 5.81$, $p < .001$, $\eta_p^2 = .124$, Fig. S2C/F), thus replicating the results of Experimental 1 and setting the stage for fMRI analysis. As in Experiment 1, the conflict similarity modulation effect remained significant after regressing out the influence of physical proximity between the stimuli of consecutive trials (Note S2).

Brain activations modulated by conflict type with univariate analyses

In the fMRI analysis, we first replicated the classic congruency effect by searching for brain regions showing higher univariate activation in incongruent than congruent conditions (GLM1, see Methods). Consistent with the literature (Botvinick et al., 2004; Fu et al., 2022), this effect

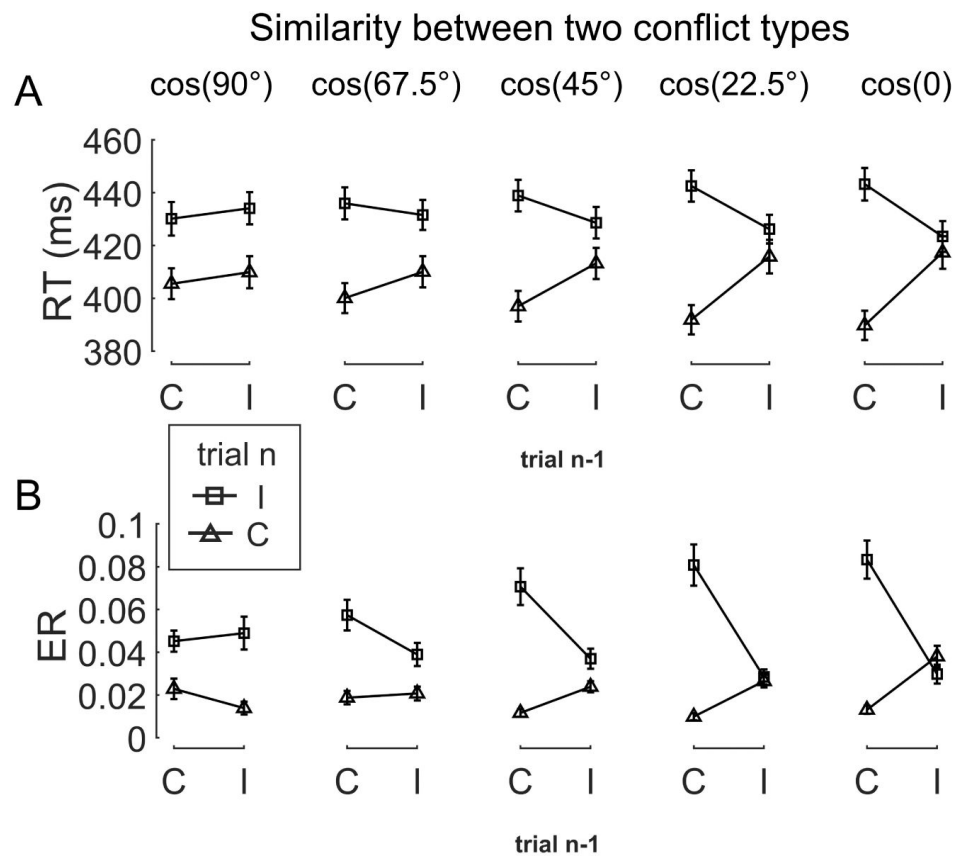


Fig. 2.

The conflict similarity modulation on the behavioral CSE in Experiment 1.

(A) RT and (B) ER are plotted as a function of congruency types on trial n-1 and trial n. Each column shows one similarity level, as indicated by the defined angular difference between two conflict types. Error bars are standard errors. C = congruent; I = incongruent; RT = reaction time; ER = error rate.

was observed in the pre-supplementary motor area (pre-SMA) (**Fig. 3**, Table S1). We then tested the encoding of conflict type as a cognitive space by identifying brain regions with activation levels parametrically covarying with the coordinates (i.e., axial angle relative to the horizontal axis) in the hypothesized cognitive space. As shown in **Fig. 1B**, change in the angle corresponds to change in spatial Stroop and Simon conflicts in opposite directions. Accordingly, we found the right inferior parietal sulcus (IPS) and the right dorsomedial prefrontal cortex (dmPFC) displayed positive correlation between fMRI activation and the Simon conflict (**Fig. 3**, Fig. S3, Table S1).

To further test if the univariate results explain the conflict similarity modulation of the behavioral CSE (slope in Fig. S2C), we conducted brain-behavioral correlation analyses for regions identified above. Regions with higher spatial Stroop/Simon modulation effects were expected to trigger higher behavioral conflict similarity modulation effect on the CSE. However, none of the three regions (i.e., left MFG, right IPS and right dmPFC, **Fig. 3**) were positively correlated with the behavioral performance, all uncorrected $ps > .28$, one-tailed. In addition, since the conflict type difference covaries with the orientation of the arrow location at the individual level (e.g., the stimulus in a higher level of Simon conflict is always closer to the horizontal axis, see **Fig. 7**), the univariate modulation effects may not reflect purely conflict type difference. To further tease these factors apart, we used multivariate analyses.

Multivariate patterns of the right dlPFC encodes the conflict similarity

The hypothesis that the brain encodes conflict types in a cognitive space predicts that similar conflict types will have similar neural representations. To test this prediction, we computed the representational similarity matrix (RSM) that encoded correlations of blood-oxygen-level dependent (BOLD) signal patterns between each pair of conflict type (Stroop, St_HSm_L , St_MSm_M , St_LSm_H , and Simon, with H, M and L indicating high, medium and low, respectively, see also **Fig. 1B**) $\times$ congruency (congruent, incongruent) $\times$ arrow direction (up, down) $\times$ run $\times$ subject combinations for each of the 360 cortical regions from the Multi-Modal Parcellation (MMP) cortical atlas (Glasser et al., 2016; Jiang et al., 2020). The RSM was then submitted to a linear mixed-effect model as the dependent variable to test whether the representational similarity in each region was modulated by various experimental variables (e.g., conflict type, spatial orientation, stimulus, response, etc., see Methods). The linear mixed-effect model was used to de-correlate conflict type and spatial orientation leveraging the between-subject manipulation of stimulus locations (**Fig. 7** and **Fig. 8**).

To validate this method, we applied this analysis to test the effects of response/stimulus features and found that representational similarity of the BOLD signal significantly covaried with whether two response/spatial location/arrow directions were the same most strongly in bilateral motor/visual/somatosensory areas, respectively (Fig. S5). We then identified the cortical regions encoding conflict type as a cognitive space by testing whether their RSMs can be explained by the similarity between conflict types. Specifically, we applied three independent criteria: (1) the cortical regions should exhibit a statistically significant positive conflict similarity effect on the RSM; (2) the conflict similarity effect should be stronger in incongruent than congruent trials to reflect flexible adjustment of cognitive control demand when the conflict is present; and (3) the conflict similarity effect should be positively correlated with the behavioral conflict similarity modulation effect on the CSE (see *Behavioral Results* of Experiment 2). The first criterion revealed several cortical regions encoding the conflict similarity, including the Brodmann 8C area (a subregion of dlPFC) (Glasser et al., 2016) and a47r in the right hemisphere, and the superior frontal language (SFL) area, 6r, 7Am, 24dd, and ventromedial visual area 1 (VMV1) areas in the left hemisphere (Bonferroni corrected $ps < 0.0001$, one-tailed, **Fig. 4A**). We next tested whether these regions were related to cognitive control by comparing the strength of conflict similarity effect between incongruent and congruent conditions (criterion 2). Results revealed that the left SFL, left VMV1, and right 8C met this criterion, Bonferroni corrected $ps < .05$, one-tailed, suggesting that the

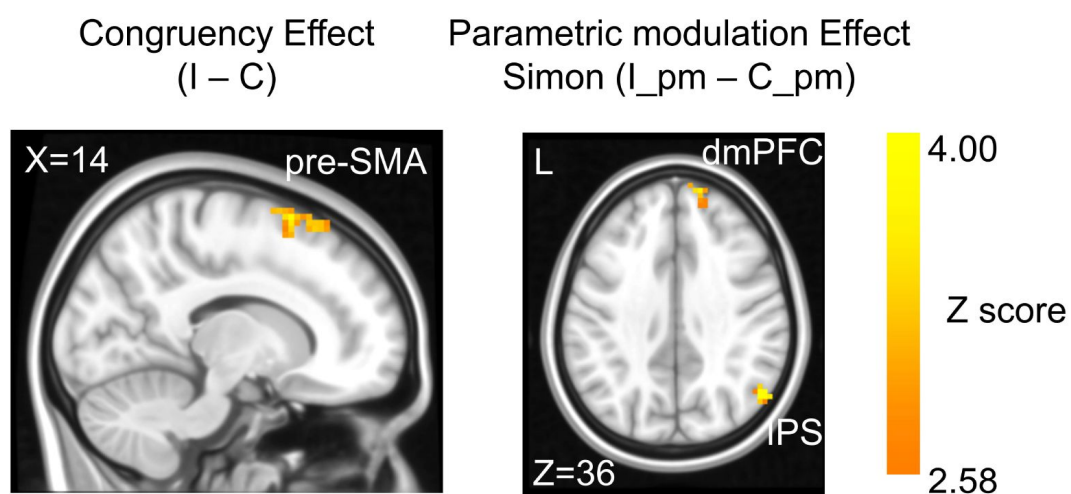


Fig. 3.

The congruency effect and parametric modulation effect detected by uni-voxel analyses.

Results displayed are probabilistic TFCE enhanced and thresholded with voxel-wise $p < .001$ and cluster-wise $p < .05$, both one-tailed. The congruency effect denotes the higher activation in incongruent than congruent condition (left panel). The positive parametric modulation effect (I_pm – C_pm) denotes the higher activation when the conflict type contained a higher ratio of Simon conflict component (right panel). I = incongruent; C = congruent; pm = parametric modulator.

representation of conflict type was strengthened when the conflict was present (e.g., **Fig. 4D** and **Fig. S4**). The inter-subject brain-behavioral correlation analysis (criterion 3) showed that the strength of conflict similarity effect on RSM scaled with the modulation of conflict similarity on the CSE (slope in **Fig. S2C**) in right 8C ($r = .52$, Bonferroni corrected $p = .002$, one-tailed, **Fig. 4C**) but not in the left SFL and VMV1 (all Bonferroni corrected $ps > .05$, one-tailed). These results are listed in **Table 1**. In addition, we did not find evidence supporting the encoding of congruency in the right 8C area (see Note S6), suggesting that the right 8C area specifically represents conflict similarity.

To examine if the right 8C specifically encodes the cognitive space rather than the domain-general or domain-specific organizations, we tested two additional models (see Methods). Model comparison showed a lower BIC in the Cognitive-Space model (BIC = 5377094) than the Domain-General (BIC = 537127) or Domain-Specific (BIC = 537127) models. Further analysis showed the dimensionality of the representation in the right 8C was 1.19, suggesting the cognitive space was close to 1D. Moreover, we replicated the results with only incongruent trials, considering that the pattern of conflict representations is more manifested when the conflict is present (i.e., on incongruent trials) than not (i.e., on congruent trials). We found a poorer fitting in Domain-general (BIC = 1344129), and Domain-Specific (BIC = 1344129) models than the Cognitive-Space model (BIC = 1344104). These results indicate that the right 8C encodes an integrated cognitive space for resolving Stroop and Simon conflicts. The more detailed model comparison results are listed in **Table 2**.

In sum, we found converging evidence supporting that the right dlPFC (8C area) encoded conflict similarity parametrically, which further supports the hypothesis that conflict types are represented in a cognitive space.

Multivariate patterns of visual and oculomotor areas encode stimulus orientation

To tease apart the representation of conflict type from that of perceptual information, we tested the modulation of the spatial orientations of stimulus locations on RSM using the aforementioned RSA. We also applied three independent criteria: (1) the cortical regions should exhibit a statistically significant orientation effect on the RSM; (2) the conflict similarity effect should be stronger in incongruent than congruent trials; and (3) the orientation effect should not interact with the CSE, since the orientation effect was dissociated from the conflict similarity effect and was not expected to influence cognitive control. We observed increasing fMRI representational similarity between trials with more similar orientations of stimulus location in the occipital cortex, such as right V1, right V2, right V4, and right lateral occipital 2 (LO2) areas (Bonferroni corrected $ps < 0.0001$). We also found the same effect in the oculomotor related region, i.e., the left frontal eye field (FEF), and other regions including the right 5m, left 31pv and right parietal area F (PF) (**Fig. 5A**). Then we tested if any of these brain regions were related to the conflict representation by comparing their encoding strength between incongruent and congruent conditions. Results showed that the right V1, right V2, left FEF, and right PF encoded stronger orientation effect in the incongruent than the congruent condition, Bonferroni corrected $ps < .05$, one-tailed (**Table 1**, **Fig. 5B**). We then tested if any of these regions was related to the behavioral performance, and results showed that none of them positively correlated with the behavioral conflict similarity modulation effect, all uncorrected $ps > .45$, one-tailed. Thus all regions are consistent with the criterion 3. Like the right 8C area, none of the reported areas directly encoded congruency (see Note S6). Taken together, we found that the visual and oculomotor regions encoded orientations of stimulus location in a continuous manner and that the encoding strength was stronger when the conflict was present.

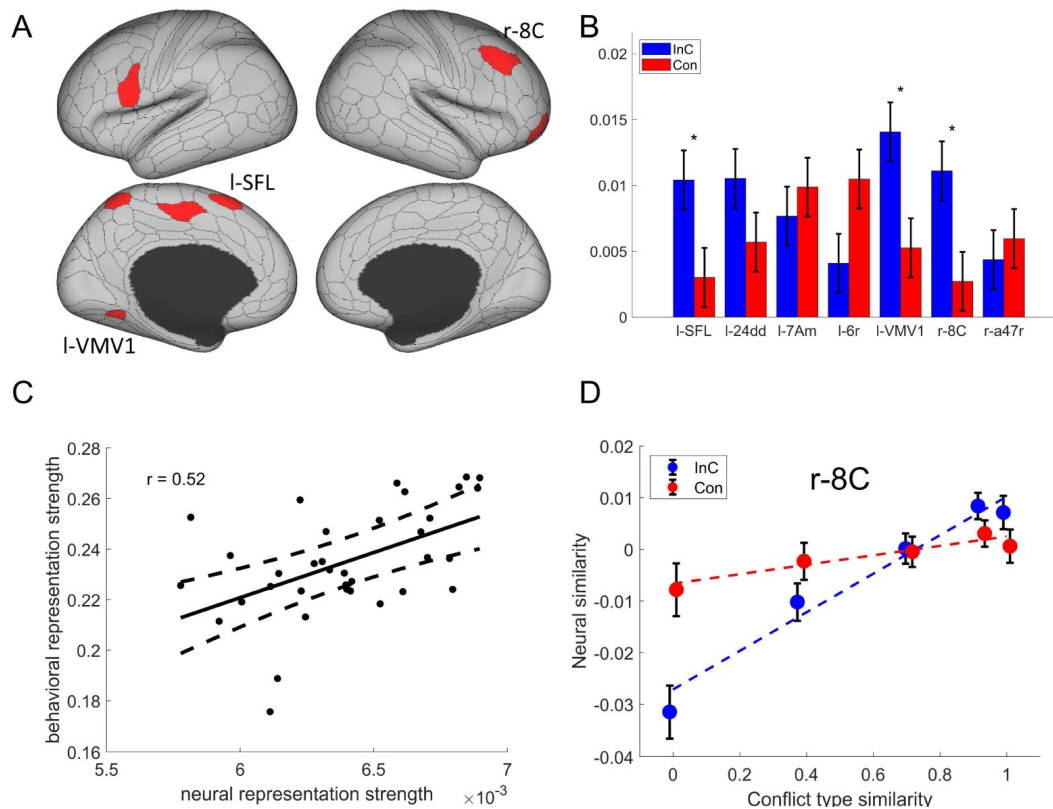


Fig. 4.

The conflict type effect.

(A) Brain regions surviving the Bonferroni correction ($p < 0.0001$) across the 360 regions (criterion 1). Labeled regions are those meeting the criterion 2. (B) Different encoding of conflict type in the incongruent with congruent conditions. * Bonferroni corrected $p < .05$. (C) The brain-behavior correlation of the right 8C (criterion 3). The x-axis shows the beta coefficient of the conflict type effect from the RSA, and the y-axis shows the beta coefficient obtained from the behavioral linear model using the conflict similarity to predict the CSE in Experiment 2. (D) Illustration of the different encoding strength of conflict type similarity in incongruent versus congruent conditions of right 8C. The y-axis is derived from the z-scored Pearson correlation coefficient, consistent with the RSA methodology. See Fig. S4B for a plot with the raw Pearson correlation measurement. l = left; r = right.

Region name	Criterion 1			Criterion 2		Criterion 3	
	<i>t</i>	$\beta \pm \text{SD}$	<i>p</i>	<i>t</i>	<i>p</i>	<i>r</i>	<i>p</i>
<i>Conflict type effect</i>							
left SFL	5.12	0.0008 ± 0.0002	1.5×10 ⁻⁷	2.45	.007	.35	.021
left 24dd	6.61	0.0012 ± 0.0002	1.9×10 ⁻¹¹	1.61	.053	-.04	.580
left 7Am	7.40	0.0015 ± 0.0002	6.6×10 ⁻¹⁴	-0.75	.772	.04	.418
left 6r	5.90	0.0012 ± 0.0002	1.8×10 ⁻⁹	-1.94	.974	-.11	.736
left VMV1	7.84	0.0017 ± 0.0002	2.2×10 ⁻¹⁵	2.83	.002	-.27	.940
right 8C*	5.60	0.0011 ± 0.0002	1.1×10 ⁻⁸	2.46	.007	.52	.001
right a47r	5.04	0.0007 ± 0.0001	2.4×10 ⁻⁷	-0.68	.753	-.04	.589
<i>Orientation effect</i>							
left FEF*	5.17	0.0011 ± 0.0002	1.2×10 ⁻⁷	2.73	.003	.02	.456
left 3lpv	6.37	0.0019 ± 0.0003	9.6×10 ⁻¹¹	-0.48	.686	.12	.247
right V1*	5.59	0.0007 ± 0.0001	1.1×10 ⁻⁸	2.91	.002	-.08	.671
right V2*	5.09	0.0015 ± 0.0003	1.8×10 ⁻⁷	3.19	.001	.02	.466
right V4	5.02	0.0015 ± 0.0003	2.6×10 ⁻⁷	0.56	.289	-.13	.777
right LO2	5.95	0.0020 ± 0.0003	1.3×10 ⁻⁹	-1.81	.965	.27	.062
right 5m	5.24	0.0011 ± 0.0002	8.1×10 ⁻⁸	-2.12	.983	-.10	.707
right PF*	5.57	0.0009 ± 0.0002	1.2×10 ⁻⁸	3.08	.001	-.01	.523

Note. all *p* values listed are 1-tailed and uncorrected. The * denotes the regions meeting all three criteria for each effect.

Table 1.

Summary statistics of the cross-subject RSA for regions showing conflict type and orientation effects identified by the three criteria.

Model name	Regressor	Full RSM			RSM_I		
		<i>t</i>	<i>p</i>	BIC	<i>t</i>	<i>p</i>	BIC
Cognitive-Space	similarity	5.60	1.07×10 ⁻⁸	5377094	4.97	3.37×10 ⁻⁷	1344105
Domain-General	similarity	-0.01	.503	5377127	0.57	.285	1344129
Domain-Specific	similarity	0.84	.201	5377127	0.28	.390	1344129

Table 2.

Model comparison results of the right 8C. RSM_I shows results using incongruent trials only.

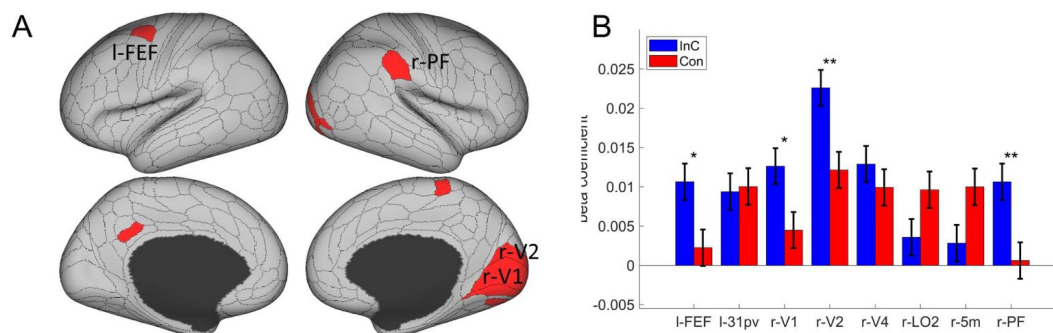


Fig. 5.

The orientation effect.

(A) Brain regions surviving the Bonferroni correction ($p < 0.0001$) across the 360 regions (criterion 1). Labeled regions are those meeting the criterion 2. (B) Different encoding of orientation in the incongruent with congruent conditions. * Bonferroni corrected $p < .05$, ** Bonferroni corrected $p < .01$.

We hypothesize that the overlapping spatial information of orientation may have facilitated the encoding of conflict types. To explore the relation between conflict type and orientation representations, we conducted representational connectivity (i.e., the similarity between two within-subject RSMs of two regions) (Kriegeskorte et al., 2008) analyses and found that among the orientation effect regions, the right V1 and right V2 showed significant representational connectivity with the right 8C compared to the controlled regions (including those encoding orientation effect but not showing larger encoding strength in incongruent than congruent conditions, as well as eight other regions encoding none of our defined effects in the main RSA, see Methods). Compared with the largest connectivity strength in the controlled regions (i.e., the right V4, $\beta = 0.7963 \pm 0.0314$), we found higher connectivity in the right V1, $\beta = 0.8633 \pm 0.0325$, and right V2, $\beta = 0.8773 \pm 0.0334$, (Fig. S6).

Discussion

Understanding how different types of conflicts are resolved is essential to answer how cognitive control achieves adaptive behavior. However, the dichotomy between domain-general and/or domain-specific processes presents a dilemma (Braem et al., 2014; Egner, 2008). Reconciliation of the two views also suffers from the inability to fully address how highly variant conflicts can be resolved by a limited set of cognitive control processes. In this study, we hypothesized that this issue can be addressed if conflicts are organized as a cognitive space. Leveraging the well-known dissociation between the spatial Stroop and Simon conflicts (Li et al., 2014; Liu et al., 2010; Wang et al., 2014), we designed five conflict types that are systematically different from each other. The cognitive space hypothesis predicted that the representational proximity/distance between two conflict types scales with their similarities/dissimilarities, which was tested at both behavioral and neural levels.

Behaviorally, we found that the CSEs were linearly modulated by conflict similarity between consecutive trials, replicating and extending our previous study (Yang et al., 2021). BOLD activity patterns in the right dlPFC further showed that the representational similarity between conflict types was modulated by their conflict similarity, and that strength of the modulation was positively associated with the modulation of conflict similarity on the behavioral CSE. We also observed that activity in two brain regions (right IPS and right dlPFC) was parametrically modulated by the conflict type difference, though they did not directly explain the behavioral results. Additionally, we found that the visual regions encoded the spatial orientation of the stimulus location, which might provide the essential concrete information to determine the conflict type. Together, these results suggest that conflicts may be organized in a cognitive space that enables a limited set of cognitive control processes to resolve numerous types of conflicts.

Conventionally, the domain-general view of control suggests a common representation for different types of conflicts (Fig. 6, left), while the domain-specific view suggests dissociated representations for different types (Fig. 6, right). Previous research on this topic often adopts a binary manipulation of conflicts (Braem et al., 2014) (i.e., each domain only has one conflict type) and gathered evidence for the domain-general/specific view with presence/absence of CSE, respectively. Here, we parametrically manipulated the similarity of conflict types and found the CSE systematically vary with conflict similarity level, demonstrating that cognitive control is neither purely domain-general nor purely domain-specific, but can be reconciled as a cognitive space (Bellmund et al., 2018) (Fig. 6, middle). The model comparison analysis also showed that the Cognitive-Space model explained the representation in right DLPFC better than the Domain-General or Domain-Specific models. Specifically, the cognitive space provides a solution to use a single cognitive space organization to encode different types of conflicts that are close (domain-general) or distant (domain-specific) to each other. It also shows the potential for how large numbers of conflict types can be coded using limited resources (i.e., as different points in a low-dimensional cognitive space), as suggested by its out-of-sample predictability. Moreover, geometry

can also emerge in the cognitive space (Fu et al., 2022), which will allow for decomposition of a conflict type (e.g., how much conflict in each of the dimensions in the cognitive space) so that it can be mapped into the limited set of cognitive control processes. Such geometry enables fast learning of cognitive control settings from similar conflict types by providing a measure of similarity (e.g., as distance in space).

If the dimensionality of the cognitive space of conflict is extremely high, the cognitive space solution would suffer the same criticism as the domain-specificity theory. We argue that the dimensionality is manageable for the human brain, as task information unrelated to differentiating conflicts can be removed. For example, the Simon conflict can be represented in a space consisting of spatial location, stimulus information and responses. Thus, the dimensionality of the cognitive space of conflicts should not exceed the number of represented features. The dimensionality can be further reduced, as humans selectively represent a small number of features when learning task representations (e.g., spatial information is reduced to the horizontal dimension from the 3D space we live in) (Niv, 2019). Consistently, we observed a low dimensional (1.19D) space representing the five conflict types. This is expected since the only manipulated variable is the angular distance between conflict types. The reduced dimensionality does not only require less effort to represent the conflict, but also facilitates generalization of cognitive control settings among different conflict types (Badre et al., 2021).

Although our finding of cognitive space in the right dlPFC differs from other cognitive space studies (Constantinescu et al., 2016; Park et al., 2020; Schuck et al., 2016) that highlighted the orbitofrontal and hippocampus regions, it is consistent with the cognitive control literature. The prefrontal cortex has long been believed to be a key region of cognitive control representation (Mansouri et al., 2007; Miller & Cohen, 2001; Milner, 1963) and is widely engaged in multiple task demands (Cole et al., 2013; Duncan, 2010). However, it is not until recently that the multivariate representation in this region has been examined. For instance, Vaidya et al. (2021) reported that frontal regions presented latent states that are organized hierarchically. Freund et al. (2021) showed that dlPFC encoded the target and congruency in a typical color-word Stroop task. Taken together, we suggest that the right dlPFC might flexibly encode a variety of cognitive spaces to meet the dynamic task demands. In addition, we found no such representation in the left dlPFC (Note S8), indicating a possible lateralization. Previous studies showed that the left dlPFC was related to the expectancy-related attentional set up-regulation, while the right dlPFC was related to the online adjustment of control (Friebs et al., 2020; Vanderhasselt et al., 2009), which is consistent with our findings. Moreover, the right PFC also represents a composition of single rules (Reverberi et al., 2012), which may explain how the spatial Stroop and Simon types can be jointly encoded in a single space.

Previous researchers have proposed an “expected value of control (EVC)” theory, which posits that the brain can evaluate the cost and benefit associated with executing control for a demanding task, such as the conflict task, and specify the optimal control strength (Shenhav et al., 2013). For instance, Grahek et al. (2022) found that more frequently switching goals when doing a Stroop task were achieved by adjusting smaller control intensity. Our work complements the EVC theory by further investigating the neural representation of different conflict conditions and how these representations can be evaluated to facilitate conflict resolution. We found that different conflict conditions could be efficiently represented in a cognitive space encoded by the right dlPFC, and participants with stronger cognitive space representation have also adjusted their conflict control to a greater extent based on the conflict similarity (Fig 4C). The finding suggests that the cognitive space organization of conflicts guides cognitive control to adjust behavior. Previous studies have shown that participants may adopt different strategies to represent a task, with the model-based strategies benefitting goal-related behaviors more than the model-free strategies (Rmus et al., 2022). Similarly, we propose that cognitive space could serve as a mental model to assist fast learning and efficient organization of cognitive control settings. Specifically, the cognitive space representation may provide a principle for how our brain evaluates the expected

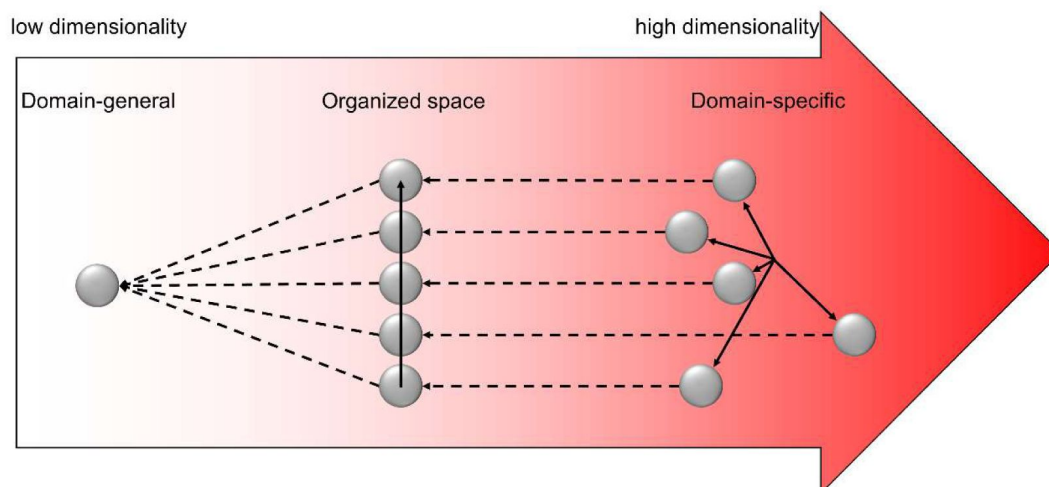


Fig. 6.

Illustration of the hypothesized dimensionalities of different representations.

The shade of the red color indicates the degree of dimensionality (i.e., how many dimensions are needed to represent different states). The dimensionality of domain-general representation is extremely low, with all representations compressed to one dot. The dimensionality of domain-specific representation is extremely high, with each control state encoded in a unique and orthogonal dimension. The dimensionality of the organized representation is modest, enabling distant states to be separated but also allowing close states to share representations. The solid arrows show the axes of different dimensions. The dashed arrows indicate how the representational dimensionality can be reduced by projecting the independent dimensions to a common dimension.

cost of switching and the benefit of generalization between states and selects the path with the best cost-benefit tradeoff (Abrahamse et al., 2016; Shenhav et al., 2013). The proximity between two states in cognitive space could reflect both the expected cognitive demand required to transition and the useful mechanisms to adapt from. The closer the two conditions are in cognitive space, the lower the expected switching cost and the higher the generalizability when transitioning between them. With the organization of a cognitive space, a new conflict can be quickly assigned a location in the cognitive space, which will facilitate the development of cognitive control settings for this conflict by interpolating nearby conflicts and/or projecting the location to axes representing different cognitive control processes, thus leading to a stronger CSE when following a more similar conflict condition. On the other hand, without a cognitive space, there would be no measure of similarity between conflicts on different trials, hence limiting the ability of fast learning of cognitive control setting from similar trials.

The cognitive space in the right dlPFC appears to be an abstraction of concrete information from the visual regions. We found that the right V1 and V2 encoded the spatial orientation of the target location (Fig. 5) and showed strong representational connectivity with the right dlPFC (Fig. S6), suggesting that there might be information exchange between these regions. We speculate that the representation of spatial orientation may have provided the essential perceptual information to determine the conflict type (Fig. 1) and thus served as the critical input for the cognitive space. The conflict type representation further incorporates the stimulus-response mapping rules to the spatial orientation representation, so that vertically symmetric orientations can be recognized as the same conflict type (Fig. 7). In other words, the representation of conflict type involves the compression of perceptual information (Flesch et al., 2022), which is consistent with the idea of a low-dimensional representation of cognitive control (Badre et al., 2021; MacDowell et al., 2022). The compression and abstraction processes might be why the frontoparietal regions are the top of hierarchy of information processing (Gilbert & Li, 2013) and why the frontoparietal regions are widely engaged in multiple task demands (Duncan, 2013).

Although the spatial orientation information in our design could be helpful to the construction of cognitive space, the cognitive space itself was independent of the stimulus-level representation of the task. We found the conflict similarity modulation on CSE did not change with more training (see Note S3), indicating that the cognitive space did not depend on strategies that could be learned through training. Instead, the cognitive space should be determined by the intrinsic similarity structure of the task design. For example, a previous study (Freitas & Clark, 2015) has found that the CSE across different versions of spatial Stroop and flanker tasks was stronger than that across either of the two conflicts and Simon. In their designs, the stimulus similarity was controlled at the same level, so the difference in CSE was only attributable to the similar dimensional overlap between Stroop and flanker tasks, in contrast to the Simon task. Furthermore, recent studies showed that the cognitive space generally exists to represent structured latent states (e.g., Vaidya & Badre, 2022), mental strategy cost (Grahek et al., 2022), and social hierarchies (Park et al., 2020). Therefore, cognitive space is likely a universal strategy that can be applied to different scenarios.

With conventional univariate analyses, we observed that the overall congruency effect was located at the medial frontal region (i.e., pre-SMA), which is consistent with previous studies (Botvinick et al., 2004; Fu et al., 2022). Beyond that, we also found regions that can be parametrically modulated by conflict type difference, including right IPS, right dlPFC (modulated by Simon difference). The right lateralization of these regions is consistent with a previous finding (Li et al., 2017). The parametric encoding of conflict also mirrors prior research showing the parametric encoding of task demands (Dagher et al., 1999; Ritz & Shenhav, 2023). The scaling of brain activities based on conflict difference is potentially important to the representational organization of different types of conflicts. However, we didn't observe their brain-behavioral relevance. One possible reason is that the conflict (dis)similarity is a combination of (dis)similarity of spatial Stroop and Simon conflicts, but each univariate region only reflects difference along a

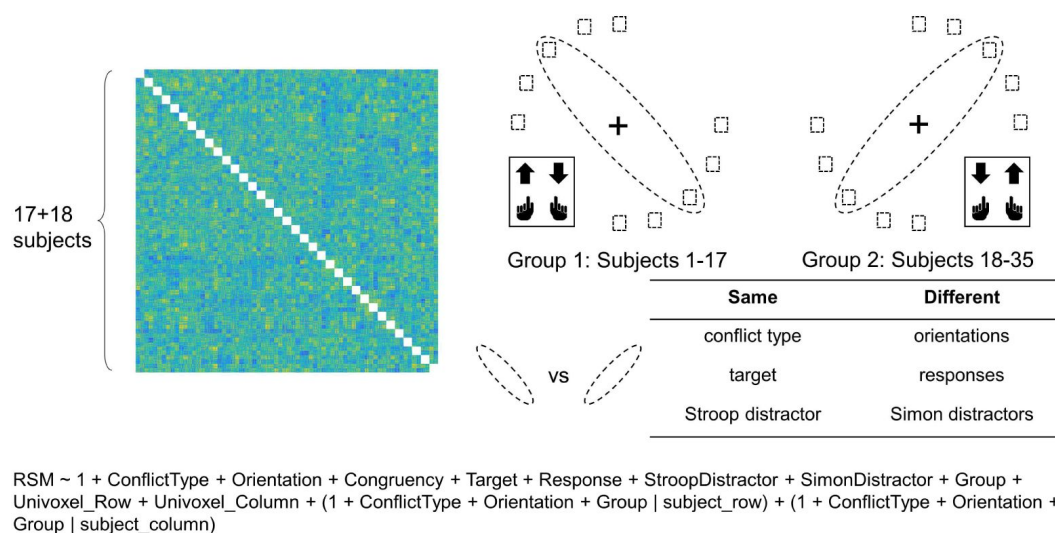


Fig. 7.

The cross-subject RSA model and the rationale. The RSM is calculated as the Pearson's correlation between each pair of conditions across the 35 subjects. For 17 subjects, the stimuli were displayed on the top-left and bottom-right quadrants, and they were asked to respond with left hand to the upward arrow and right hand to the downward arrow. For the other 18 subjects, the stimuli were displayed on the top-right and bottom-left quadrants, and they were asked to respond with left hand to the downward arrow and right hand to the upward arrow. Within each subject, the conflict type and orientation regressors were perfectly covaried. For instance, the same conflict type will always be on the same orientation. To de-correlate conflict type and orientation effects, we conducted the RSA across subjects from different groups. For example, the dashed ellipses highlight the conditions that are orthogonal to each other on the orientation representation, response, and Simon distractor, when their conflict type, target and spatial Stroop distractor are the same. The dashed boxes show the possible target locations for different conditions. RSM = representational similarity matrix.

single conflict domain. Also likely, the representational geometry is more of a multivariate problem than what univariate activities can capture (Freund, Etzel, et al., 2021 [DOI](#)). Future studies may adopt approaches such as repetition suppression induced fMRI adaptation (Badre et al., 2021 [DOI](#)) to test the role of univariate activities in task representations.

Recently an interesting debate has arisen concerning whether cognitive control should be considered as a process or a representation (Freund, Etzel, et al., 2021 [DOI](#)). Traditionally, cognitive control has been predominantly viewed as a process. However, the study of its representation has gained more and more attention. While it may not be as straightforward as the visual representation (e.g., creating a mental image from a real image in the visual area), cognitive control can have its own form of representation. An influential theory, Marr's (1982) [DOI](#) three-level model proposed that representation serves as the algorithm of the process to achieve a goal based on the input. In other words, representation can encompass a dynamic process rather than being limited to static stimuli. Building on this perspective, we posit that the representation of cognitive control consists of an array of dynamic representations embedded within the overall process. A similar idea has been proposed that the representation of task profiles can be progressively constructed with time in the brain (Kikumoto & Mayr, 2020 [DOI](#)). Moreover, we anticipate that the representation of cognitive space is most prominently involved at two critical stages to guide the transference of behavioral CSE. The first stage involves the evaluation of control demands, where the representational distance/similarity between previous and current trials influences the adjustment of cognitive control. The second stage pertains to control execution, where the switch from one control state to another follows a path within the cognitive space. However, we were unable to fully distinguish between these two stages due to the low temporal resolution of fMRI signals in our study. Future research seeking to delve deeper into this question may benefit from methodologies with higher temporal resolutions, such as EEG and MEG.

Several interesting questions remains to be answered. For example, is the dimension of the unified space across conflict-inducing tasks solely determined by the number of conflict sources? Is this unified space adaptively adjusted within the same brain region? Can we effectively map any sources of conflict with completely different stimuli into a single space? Does the cognitive space vary from population to population, such as between the normal people and patients?

Methodological implications. The use of an experimental paradigm that permits parametric manipulation of conflict similarity provides a way to systematically investigate the organization of cognitive control, as well as its influence on adaptive behaviors. This approach extends traditional paradigms, such as the multi-source interference task (Fu et al., 2022 [DOI](#)), color Stroop-Simon task (Liu et al., 2010 [DOI](#)) and similar paradigms that do not afford a quantifiable metric of conflict source similarity. Moreover, the cross-subject RSA provides high sensitivity to the variables of interest and the ability to separate confounding factors. For instance, in addition to dissociating conflict type from orientation, we dissociated target from response, and spatial Stroop distractor from Simon distractor. We further showed cognitive control can both enhance the target representation and suppress the distractor representation (Note S10, Fig. S7), which is in line with previous studies (Polk et al., 2008 [DOI](#); Ritz & Shenhav, 2022 [DOI](#)).

A few limitations of this study need to be noted. To parametrically manipulate the conflict similarity levels, we adopted the spatial Stroop-Simon paradigm that enables parametrical combinations of spatial Stroop and Simon conflicts. However, since this paradigm is a two-alternative forced choice design, the behavioral CSE is not a pure measure of adjusted control but could be partly confounded by bottom-up factors such as feature integration (Hommel et al., 2004 [DOI](#)). Future studies may replicate our findings with a multiple-choice design (including more varied stimulus sets, locations and responses) with confound-free trial sequences (Braem et al., 2019 [DOI](#)). Another limitation is that in our design, the spatial Stroop and Simon effects are highly anticorrelated. This constraint may make the five conflict types represented in a unidimensional space (e.g., a circle) embedded in a 2D space. Future studies may test the 2D cognitive space with

fully independent conditions. A possible improvement to our current design would be to include left, right, up, and down arrows represented in a grid formation across four spatially separate quadrants, with each arrow mapped to its own response button. However, one potential confounding factor would be that these conditions have different levels of difficulty (i.e., different magnitude of conflict), which may affect the CSE results and their representational similarity. Additionally, our study is not a comprehensive test of the cognitive space hypothesis but aimed primarily to provide original evidence for the geometry of cognitive space in representing conflict information in cognitive control. Future research should examine other aspects of the cognitive space such as its dimensionality, its applicability to other conflict tasks such as Eriksen Flanker task, and its relevance to other cognitive abilities, such as cognitive flexibility and learning.

In sum, we showed that the cognitive control can be parametrically encoded in the right dlPFC and guides cognitive control to adjust goal-directed behavior. This finding suggests that different cognitive control states may be encoded in an abstract cognitive space, which reconciles the long-standing debate between the domain-general and domain-specific views of cognitive control and provides a parsimonious and more broadly applicable framework for understanding how our brains efficiently and flexibly represents multiple task settings.

Materials and Methods

Subjects

In Experiment 1, we enrolled thirty-three college students (19–28 years old, average of 21.5 ± 2.3 years old; 19 males). In Experiment 2, thirty-six college students were recruited, and one subject was excluded due to not following task instructions. The final sample of Experiment 2 consisted of thirty-five participants (19–29 years old, average of 22.3 ± 2.5 years old; 17 males). The sample sizes were determined based on our previous study (Yang et al., 2021 [DOI](#)). All participants reported no history of psychiatric or neurological disorders and were right-handed, with normal or corrected-to-normal vision. The experiments were approved by the Institutional Review Board of the Institute of Psychology, Chinese Academy of Science. Informed consent was obtained from all subjects.

Method Details

Experiment 1

Experimental Design

We adopted a modified spatial Stroop-Simon task (Yang et al., 2021 [DOI](#)) (**Fig. 1** [DOI](#)). The task was programmed with the E-prime 2.0 (Psychological Software Tools, Inc.). The stimulus was an upward or downward black arrow (visual angle of $\sim 1^\circ$) displayed on a 17-inch LCD monitor with a viewing distance of ~ 60 cm. The arrow appeared inside a grey square at one of ten locations with the same distance from the center of the screen, including two horizontal (left and right), two vertical (top and bottom), and six corner (orientations of 22.5° , 45° and 67.5°) locations. The distance from the arrow to the screen center was approximately 3° . To dissociate orientation of stimulus locations and conflict types (see below), participants were randomly assigned to two sets of stimulus locations (one included top-right and bottom-left quadrants, and the other included top-left and bottom-right quadrants).

Each trial started with a fixation cross displayed in the center for 100–300 ms, followed by the arrow for 600 ms and another fixation cross for 1100–1300 ms (the total trial length was fixed at 2000 ms). Participants were instructed to respond to the pointing direction of the arrow by pressing a left or right button and to ignore its location. The mapping between the arrow

orientations and the response buttons was counterbalanced across participants. The task design introduced two possible sources of conflicts: on one hand, the direction of the arrow is either congruent or incongruent with the vertical location of the arrow, thus introducing a spatial Stroop conflict (Lu & Proctor, 1995 [↗](#); MacLeod, 1991 [↗](#)), which contains the dimensional overlap between task-relevant stimulus and task-irrelevant stimulus (Kornblum et al., 1990 [↗](#)); on the other hand, the response (left or right button) is either congruent or incongruent with the horizontal location of the arrow, thus introducing a Simon conflict (Lu & Proctor, 1995 [↗](#); Simon & Small, 1969 [↗](#)), which contains the dimensional overlap between task-irrelevant stimulus and response (Kornblum et al., 1990 [↗](#)). Therefore, the five polar orientations of the stimulus location (from 0 to 90°) defined five unique combinations of spatial Stroop and Simon conflicts, with more similar orientations having more similar composition of conflicts. More generally, the spatial orientation of the arrow location relative to the center of the screen forms a cognitive space of different blending of spatial Stroop and Simon conflicts.

The formal task consisted of 30 runs of 101 trials each, divided into three sessions of ten runs each. The participants completed one session each time and all three sessions within one week. Before each session, the participants performed training blocks of 20 trials repeatedly until the accuracy reached 90% in the most recent block. The trial sequences of the formal task were pseudo-randomly generated to ensure that each of the task conditions and their transitions occurred with equal number of trials.

Experiment 2

Experimental Design

The apparatus, stimuli and procedure were identical to Experiment 1 except for the changes below. The stimuli were back projected onto a screen (with viewing angle being $\sim 3.9^\circ$ between the arrow and the center of the screen) behind the subject and viewed via a surface mirror mounted onto the head coil. Due to the time constraints of fMRI scanning, the trial numbers decreased to a total of 340, divided into two runs with 170 trials each. To obtain a better hemodynamic model fitting, we generated two pseudo-random sequences optimized with a genetic algorithm (Wager & Nichols, 2003 [↗](#)) conducted by the NeuroDesign package (Durnez et al., 2018 [↗](#)) (see Note S3 for more detail). In addition, we added 6 seconds of fixation before each run to allow the stabilization of the hemodynamic signal, and 20 seconds after each run to allow the signal to drop to the baseline.

Before scanning, participants performed two practice sessions. The first one contained 10 trials of center-displayed arrow and the second one contained 32 trials using the same design as the main task. They repeated both sessions until their performance accuracy for each session reached 90%, after which the scanning began.

fMRI Image acquisition and preprocessing

Functional imaging was performed on a 3T GE scanner (Discovery MR750) using echo-planar imaging (EPI) sensitive to BOLD contrast [in-plane resolution of $3.5 \times 3.5 \text{ mm}^2$, 64×64 matrix, 37 slices with a thickness of 3.5 mm and no interslice skip, repetition time (TR) of 2000 ms, echo-time (TE) of 30 ms, and a flip angle of 90°]. In addition, a sagittal T1-weighted anatomical image was acquired as a structural reference scan, with a total of 256 slices at a thickness of 1.0 mm with no gap and an in-plane resolution of $1.0 \times 1.0 \text{ mm}^2$.

Before preprocessing, the first three volumes of the functional images were removed due to the instability of the signal at the beginning of the scan. The anatomical and functional data were preprocessed with the fMRIPrep 20.2.0 (Esteban et al., 2019 [↗](#)) (RRID:SCR_016216), which is based on Nipype 1.5.1 (Gorgolewski et al., 2011 [↗](#)) (RRID:SCR_002502). Specifically, BOLD runs were slice-time corrected using 3dTshift from AFNI 20160207 (jenkinson et al., 2002 [↗](#)) (RRID:SCR_005927).

The BOLD time-series were resampled to the MNI152NLin2009cAsym space without smoothing. For a more detailed description of preprocessing, see Note S4. After preprocessing, we resampled the functional data to a spatial resolution of $3 \times 3 \times 3 \text{ mm}^3$. All analyses were conducted in volumetric space, and surface maps are produced with Connectome Workbench (<https://www.humanconnectome.org/software/connectome-workbench>) for display purpose only.

Quantification and Statistical Analysis

Behavioral analysis

Experiment 1

RT and ER were the two dependent variables analyzed. As for RTs, we excluded the first trial of each block (0.9%, for CSE analysis only), error trials (3.8%), trials with RTs beyond three *SDs* or shorter than 200 ms (1.3%) and post-error trials (3.4%). For the ER analysis, the first trial of each block and trials after an error were excluded. To exclude the possible influence of response repetition, we centered the RT and ER data within the response repetition and response alternation conditions separately by replacing condition-specific mean with the global mean for each subject.

To examine the modulation of conflict similarity on the CSE, we organized trials based on a 5 (previous trial conflict type) $\times$ 5 (current trial conflict type) $\times$ 2 (previous trial congruency) $\times$ 2 (current trial congruency) factorial design. As conflict similarity is commutative between conflict types, we expected the previous by current trial conflict type factorial design to be a symmetrical (e.g., a conflict 1-conflict 2 sequence in theory has the same conflict similarity modulation effect as a conflict 2-conflict 1 sequence), resulting a total of 15 conditions left for the first two factors of the design (i.e., previous $\times$ current trial conflict type). For each previous $\times$ current trial conflict type condition, the conflict similarity between the two trials can be quantified as the cosine of their angular difference. In the current design, there were five possible angular difference levels (0, 22.5°, 42.5°, 67.5° and 90°, see **Fig. 1C**). We further coded the previous by current trial congruency conditions (hereafter abbreviated as CSE conditions) as CC, CI, IC and II, with the first and second letter encoding the congruency (C) or incongruency (I) on the previous and current trial, respectively. As the CSE is operationalized as the interaction between previous and current trial congruency, it can be rewritten as a contrast of (CI – CC) – (II – IC). In other words, the load of CSE on CI, CC, II and IC conditions is 1, –1, –1 and 1, respectively. To estimate the modulation of conflict similarity on the CSE, we built a regressor by calculating the Kronecker product of the conflict similarity scores of the 15 previous $\times$ current trial conflict similarity conditions and the CSE loadings of previous $\times$ current trial congruency conditions. This regressor was regressed against RT and ER data separately, which were normalized across participants and CSE conditions. The regression was performed using a linear mixed-effect model, with the intercept and the slope of the regressor for the modulation of conflict similarity on the CSE as random effects (across both participants and the four CSE conditions). As a control analysis, we built a similar two-stage model (Yang et al., 2021). In the first stage, the CSE [i.e., (CI – CC) – (II – IC)] for each of the previous $\times$ current trial conflict similarity condition was computed. In the second stage, CSE was used as the dependent variable and was predicted using conflict similarity across the 15 previous $\times$ current trial conflict type conditions. The regression was also performed using a linear mixed effect model with the intercept and the slope of the regressor for the modulation of conflict similarity on the CSE as random effects (across participants). *Experiment 2.* Behavioral data was analyzed using the same linear mixed effect model as Experiment 1, with all the CC, CI, IC and II trials as the dependent variable. In addition, to test if fMRI activity patterns may explain the behavioral representations differently in congruent and incongruent conditions, we conducted the same analysis to measure behavioral modulation of conflict similarity on the CSE using congruent (CC and IC) and incongruent (CI and II) trials separately.

Estimation of fMRI activity with univariate general linear model (GLM)

To estimate voxel-wise fMRI activity for each of the experimental conditions, the preprocessed fMRI data of each run were analyzed with the GLM. We conducted three GLMs for different purposes. GLM1 aimed to validate the design of our study by replicating the engagement of frontoparietal activities in conflict processing documented in previous studies (Jiang & Egner, 2014 [↗](#); Li et al., 2017 [↗](#)), and to explore the cognitive space related regions that were parametrically modulated by the conflict type. Preprocessed functional images were smoothed using a 6-mm FWHM Gaussian kernel. We included incongruent and congruent conditions as main regressors and appended a parametric modulator for each condition. The modulation parameters for Stroop, St_HSm_L , St_MSm_M , St_LSm_H , and Simon trials were -2, -1, 0, 1 and 2, respectively. In addition, we also added event-related nuisance regressors, including error/missed trials, outlier trials (slower than three SDs of the mean or faster than 200 ms) and trials within two TRs of significant head motion (i.e., outlier TRs, defined as standard DVARS > 1.5 or FD > 0.9 mm from previous TR) (Jiang et al., 2020 [↗](#)). On average there were 1.2 outlier TRs for each run. These regressors were convolved with a canonical hemodynamic response function (HRF) in SPM 12 (<http://www.fil.ion.ucl.ac.uk/spm> [↗](#)). We further added volume-level nuisance regressors, including the six head motion parameters, the global signal, the white matter signal, the cerebrospinal fluid signal, and outlier TRs. Low-frequency signal drifts were filtered using a cutoff period of 128 s. The two runs were regarded as different sessions and incorporated into a single GLM to get more power. This yielded two beta maps (i.e., a main effect map and a parametric modulation map) for the incongruent and congruent conditions, respectively and for each subject. At the group level, paired t-tests were conducted between incongruent and congruent conditions, one for the main effect and the other for the parametric modulation effect. Since the spatial Stroop and Simon conflicts change in the opposite direction to each other, a positive modulation effect would reflect a higher brain activation when there is more Simon conflict, and a negative modulation effect would reflect a higher brain activation for more spatial Stroop conflict. To avoid confusion, we converted the modulation effect of spatial Stroop to positive by using a contrast of $[-(I_{pm} - C_{pm})]$ throughout the results presentation. Results were corrected with the probabilistic threshold-free cluster enhancement (pTFCE) and then thresholded by 3dClustSim function in AFNI (Cox & Hyde, 1997 [↗](#)) with voxel-wise $p < .001$ and cluster-wise $p < .05$, both 1-tailed. To visualize the parametric modulation effects, we conducted a similar GLM (GLM2), except we used incongruent and congruent conditions from each conflict type as separate regressors with no parametric modulation. Then we extracted beta coefficients for each regressor and each participant with regions observed in GLM1 as regions of interest, and finally got the incongruent-congruent contrasts for each conflict type at the individual level. We reported the results in Fig. 3 [↗](#), Table S1, and Fig. S3. Visualization of the uni-voxel results was made by the MRICron (<https://www.mccauslandcenter.sc.edu/mricro/mricron/> [↗](#)).

The GLM3 aimed to prepare for the representational similarity analysis (see below). There were several differences compared to GLM1. The unsmoothed functional images after preprocessing were used. This model included 20 event-related regressors, one for each of the 5 (conflict type) $\times$ 2 (congruency condition) $\times$ 2 (arrow direction) conditions. The event-related nuisance regressors were similar to GLM1, but with additional regressors of response repetition and post-error trials to account for the nuisance inter-trial effects. To fully expand the variance, we conducted one GLM analysis for each run. After this procedure, a voxel-wise fMRI activation map was obtained per condition, run and subject.

Representational similarity analysis (RSA)

To measure the neural representation of conflict similarity, we adopted the RSA. RSAs were conducted on each of the 360 cortical regions of a volumetric version of the MMP cortical atlas (Glasser et al., 2016 [↗](#)). To de-correlate the factors of conflict type and orientation of stimulus location, we leveraged the between-subject manipulation of stimulus locations and conducted RSA

in a cross-subject fashion (**Fig. 7** and **Fig. 8**). Previous studies (e.g., J. Chen et al., 2017) have demonstrated that consistent multi-voxel activation patterns exist across individuals, and successful applications of cross-subject RSA (see review by Freund, Etzel, et al., 2021) and cross-subject decoding approaches (Jiang et al., 2016; Tusche et al., 2016) have also been reported. The beta estimates from GLM3 were noise-normalized by dividing the original beta coefficients by the square root of the covariance matrix of the error terms (Nili et al., 2014). For each cortical region, we calculated the Pearson's correlations between fMRI activity patterns for each run and each subject, yielding a $1400 (20 \text{ conditions} \times 2 \text{ runs} \times 35 \text{ participants}) \times 1400$ RSM. The correlations were calculated in a cross-voxel manner using the fMRI activation maps obtained from GLM3 described in the previous section. We excluded within-subject cells from the RSM (thus also excluding the within-run similarity as suggested by Walther et al., (2016)), and the remaining cells were converted into a vector, which was then z-transformed and submitted to a linear mixed effect model as the dependent variable. The linear mixed effect model also included regressors of conflict similarity and orientation similarity. Importantly, conflict similarity was based on how Simon and spatial Stroop conflicts are combined and hence was calculated by first rotating all subject's stimulus location to the top-right and bottom-left quadrants, whereas orientation was calculated using original stimulus locations. As a result, the regressors representing conflict similarity and orientation similarity were de-correlated. Similarity between two conditions was measured as the cosine value of the angular difference. Other regressors included a target similarity regressor (i.e., whether the arrow directions were identical), a response similarity regressor (i.e., whether the correct responses were identical); a spatial Stroop distractor regressor (i.e., vertical distance between two stimulus locations); a Simon distractor regressor (i.e., horizontal distance between two stimulus locations). Additionally, we also included a regressor denoting the similarity of Group (i.e., whether two conditions are within the same subject group, according to the stimulus-response mapping). We also added two regressors including ROI-mean fMRI activations for each condition of the pair to remove the possible uni-voxel influence on the RSM. A last term was the intercept. To control the artefact due to dependence of the correlation pairs sharing the same subject, we included crossed random effects (i.e., row-wise and column-wise random effects) for the intercept, conflict similarity, orientation and the group factors (G. Chen et al., 2017). Individual effects for each regressor were also extracted from the model for brain-behavioral correlation analyses. In brain-behavioral analyses, only the RT was used as behavioral measure to be consistent with the fMRI results, where the error trials were regressed out.

The statistical significance of these beta estimates was based on the outputs of the mixed-effect model estimated with the “fitlme” function in Matlab 2022a. Since symmetric cells from the RSM matrix were included in the mixed-effect model, we adjusted the t and p values with the true degree of freedom, which is half of the cells included minus the number of fixed regressors. Multiple comparison correction was applied with the Bonferroni approach across all cortical regions at the $p < 0.0001$ level. To test if the representation strengths are different between congruent and incongruent conditions, we also conducted the RSA using only congruent (RDM_C) and incongruent (RDM_I) trials separately. The contrast analysis was achieved by an additional model with both RDM_C and RDM_I included, adding the congruency and the interaction between conflict type (and orientation) and congruency as both fixed and random factors. The difference between incongruent and congruent representations was indicated by a significant interaction effect. To visualize the difference, we plotted the effect-related patterns (the predictor multiplied by the slope, plus the residual) as a function of the similarity levels (**Fig. 4D**), and a summary RSM for incongruent and congruent conditions based on the raw Pearson correlation coefficients, respectively (**Fig. S4**).

Model comparison and representational dimensionality

To estimate if the right 8C specifically encodes the cognitive space, rather than the domain-general or domain-specific structures, we conducted two more RSAs. We replaced the cognitive space-based conflict similarity matrix in the RSA we reported above (hereafter referred to as the

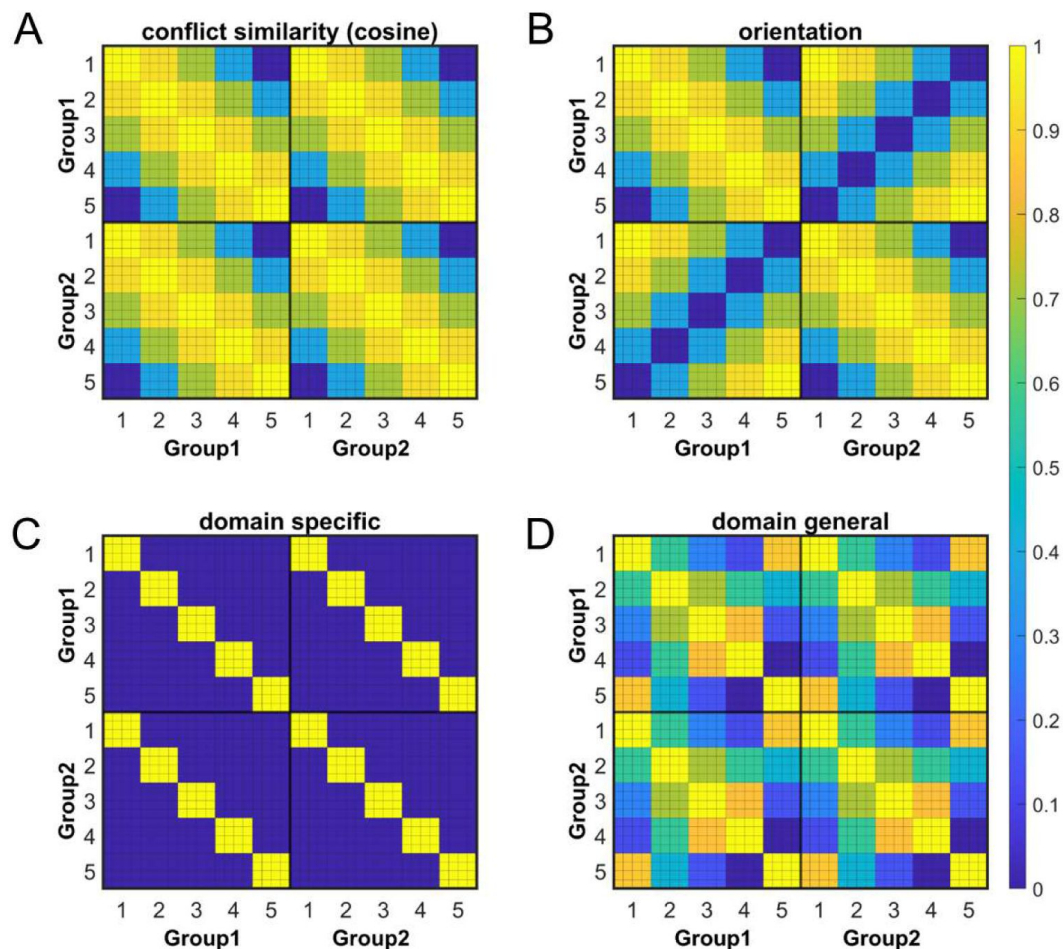


Figure 8.

Schematic of key RSMs.

(A) and (B) show the orthogonality between conflict similarity and orientation RSMs. The within-subject RSMs (e.g., Group1-Group1) for conflict similarity and orientation are all the same, but the cross-group correlations (e.g., Group2-Group1) are different. Therefore, we can separate the contribution of these two effects when including them as different regressors in the same linear regression model. (C) and (D) show the two alternative models. Like the cosine model (A), within-group trial pairs resemble between-group trial pairs in these two models. The domain-specific model is an identity matrix. The domain-general model is estimated from the absolute difference of behavioral congruency effect, but scaled to 0 (lowest similarity) – 1 (highest similarity) to aid comparison. The plotted matrices here include only one subject each from Group 1 and Group 2. Numbers 1-5 indicate the conflict type conditions, for spatial Stroop, StHSmL, StMSmM, StLSmH, and Simon, respectively. The thin lines separate four different sub-conditions, i.e., target arrow (up, down) × congruency (incongruent, congruent), within each conflict type.

Cognitive-Space model) with one of the alternative model matrices, with all other regressors equal. The domain-general model treats each conflict type as equivalent, so each two conflict types only differ in the magnitude of their conflict. Therefore, we defined the domain-general matrix as the absolute difference in their congruency effects indexed by the group-averaged RT in Experiment 2. Then the z-scored model vector was sign-flipped to reflect similarity instead of distance. The domain-specific model treats each conflict type differently, so we used a diagonal matrix, with within-conflict type similarities being 1 and all cross-conflict type similarities being 0.

Moreover, to examine if the cognitive space is driven solely by the Stroop or Simon conflicts, we tested a spatial Stroop-Only (hereafter referred to as “Stroop-Only”) and a Simon-Only model, with each conflict type projected onto the spatial Stroop (vertical) axis or Simon (horizontal) axis, respectively. The similarity between any two conflict types was defined using the overlapping of projected vectors. For example, the similarity between St_HSm_L and St_LSm_H in the Stroop-only model would be $\cos(67.5^\circ)$. We also included a model assuming the Stroop and Simon dimensions are independently represented in the brain, adding up the Stroop-Only and Simon-Only regressors (hereafter referred to as the Stroop+Simon model). We conducted similar RSAs as reported above, replacing the original conflict similarity regressor with the Stroop-Only, Simon-Only, or both regressors (for the Stroop+Simon model), and then calculated their Bayesian information criteria (BICs).

To better capture the dimensionality of the representational space, we estimated its dimensionality using the participation ratio (Ito & Murray, 2023). Since we excluded the within-subject cells from the whole RSM, the whole RSM is an incomplete matrix and could not be used. To resolve this issue, we averaged the cells corresponding to each pair of conflict types to obtain an averaged 5×5 RSM matrix, similar to the matrix shown in Fig. 1C. We then estimated the participation ratio using the formula:

$$\text{dim} = \frac{(\sum_i^m \lambda_i)^2}{\sum_i^m \lambda_i^2},$$

where λ_i is the eigenvalue of the RSM and m is the number of eigenvalues.

Representational connectivity analysis

To explore the possible relevance between the conflict type and the orientation effects, we conducted representational connectivity (Kriegeskorte et al., 2008) between regions showing evidence encoding conflict similarity and orientation similarity. We hypothesized that this relationship should exist at the within-subject level, so we conducted this analysis using within-subject RSMs excluding the diagonal. Specifically, the z-transformed RSM vector of each region were extracted and submitted to a mixed linear model, with the RSM of the conflict type region (i.e., the right 8C) as the dependent variable, and the RSM of one of the orientation regions (e.g., right V2) as the predictor. Intercept and the slope of the regressor were set as random effects at the subject level. The mixed effect model was conducted for each pair of regions, respectively. Considering there might be strong intrinsic correlations across the RSMs induced by the nuisance factors, such as the within-subject similarity, we added two sets of regions as control. First, we selected regions without showing any effects of interest (i.e., uncorrected $ps > 0.3$ for all the conflict type, orientation, congruency, target, response, spatial Stroop distractor and Simon distractor effects). Second, we selected regions of orientation effect meeting the first but not the second criterion, to account for the potential correlation between regions of the two partly orthogonal regressors (Fig. 8). Existence of representational connectivity was defined by a connectivity slope higher than 95% of the standard error above the mean of any control region.

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Reviewing Editor

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Reviewer #1 (Public Review):

People can perform a wide variety of different tasks, and a long-standing question in cognitive neuroscience is how the properties of different tasks are represented in the brain. The authors develop an interesting task that mixes two different sources of difficulty, and find that the brain appears to represent this mixture on a continuum, in the prefrontal areas involved in resolving task difficulty. While these results are interesting and in several ways compelling, they overlap with previous findings and rely on novel statistical analyses that may require further validation.

Strengths

1. The authors present an interesting and novel task for combining the contributions of stimulus-stimulus and stimulus-response conflict. While this mixture has been measured in the multi-source interference task (MSIT), this task provides a more graded mixture between these two sources of difficulty.
2. The authors do a good job triangulating regions that encoding conflict similarity, looking for the conjunction across several different measures of conflict encoding. These conflict measures use several best-practice approaches towards estimating representational similarity.
3. The authors quantify several salient alternative hypothesis, and systematically distinguish their core results from these alternatives.
4. The question that the authors tackle is important to cognitive control, and they make a solid contribution.

Concerns

1. The framing of 'infinite possible types of conflict' feels like a strawman. While they might be true across stimuli (which may motivate a feature-based account of control), the authors explore the interpolation between two stimuli. Instead, this work provides confirmatory evidence that task difficulty is represented parametrically (e.g., consistent with literatures like n-back, multiple object tracking, and random dot motion). This parametric encoding is standard in feature-based attention, and it's not clear what the cognitive map framing is contributing.
2. The representations within DLPFC appear to treat 100% Stoop and (to a lesser extent) 100% Simon differently than mixed trials. Within mixed trials, the RDM within this region don't strongly match the predictions of the conflict similarity model. It appears that there may be a more complex relationship encoded in this region.
3. To orthogonalized their variables, the authors need to employ a complex linear mixed effects analysis, with a potential influence of implementation details (e.g., high-level interactions and inflated degrees of freedom).

- <https://doi.org/10.7554/eLife.87126.3.sa1>

Reviewer #2 (Public Review):

Summary

This study examines the construct of "cognitive spaces" as they relate to neural coding schemes present in response conflict tasks. The authors use a novel experimental design in

which different types of response conflict (spatial Stroop, Simon) are parametrically manipulated. These conflict types are hypothesized to be encoded jointly, within an abstract "cognitive space", in which distances between task conditions depend only on the similarity of conflict types (i.e., where conditions with similar relative proportions of spatial-Stroop versus Simon conflicts are represented with similar activity patterns). Authors contrast such a representational scheme for conflict with several other conceptually distinct schemes, including a domain-general, domain-specific, and two task-specific schemes. The authors conduct a behavioral and fMRI study to test whether prefrontal cortex activity is correlated to one of these coding schemes. Replicating the authors' prior work, this study demonstrates that sequential behavioral adjustments (the congruency sequence effect) are modulated as a function of the similarity between conflict types. In fMRI data, univariate analyses identified activation in left prefrontal and dorsomedial frontal cortex that was modulated by the amount of Stroop or Simon conflict present, and representational similarity analyses that identified coding of conflict similarity, as predicted under the cognitive space model, in right lateral prefrontal cortex.

Strengths

This study addresses an important question regarding how conflict or difficulty might be encoded in the brain within a computationally efficient representational format. Relative to the other models reported in the paper, the evidence in support of the cognitive space model is solid. The ideas postulated by the authors are interesting and valuable ones, worthy of follow-up work that provides additional necessary scrutiny of the cognitive-space account.

Weaknesses

Future, within-subject experiments will be necessary to disentangle the cognitive space model from confounded task variables. A between-subjects manipulation of stimulus orientation/location renders the results difficult to interpret, as the source and spatial scale of the conflict encoding on cortex may differ from more rigorous (and more typical) within-subject manipulations.

Results are also difficult to interpret because Stroop and Simon conflict are confounded with each other. For interpretability, these two sources of conflict need to be manipulated orthogonally, so that each source of conflict (as well as their interaction) could be separately estimated and compared in terms of neural encoding. For example, it is therefore not clear whether the RSA results are due to encoding of only one type of conflict (Stroop or Simon), to a combination of both, and/or to interactive effects.

Finally, the motivation for the use of the term "cognitive space" to describe results is unclear. Evidence for the mere presence of a graded/parametric neural encoding (i.e., the reported conflict RSA effects) would not seem to be sufficient. Indeed, it is discussed in the manuscript that cognitive spaces/maps allow for flexibility through inference and generalization. Future work should therefore focus on linking neural conflict encoding to inference and generalization more directly.

- <https://doi.org/10.7554/eLife.87126.3.sa0>

Author Response

The following is the authors' response to the previous reviews.

Thank you and the reviewers for further providing constructive comments and suggestions on our manuscript. On behalf of all the co-authors, I have enclosed a revised version of the above referenced paper. Below, I have merged similar public reviews and recommendations (if applicable) from each reviewer and provided point-by-point responses.

Reviewer #1:

People can perform a wide variety of different tasks, and a long-standing question in cognitive neuroscience is how the properties of different tasks are represented in the brain. The authors develop an interesting task that mixes two different sources of difficulty, and find that the brain appears to represent this mixture on a continuum, in the prefrontal areas involved in resolving task difficulty. While these results are interesting and in several ways compelling, they overlap with previous findings and rely on novel statistical analyses that may require further validation.

Strengths

- 1. The authors present an interesting and novel task for combining the contributions of stimulus-stimulus and stimulus-response conflict. While this mixture has been measured in the multi-source interference task (MSIT), this task provides a more graded mixture between these two sources of difficulty.*
- 1. The authors do a good job triangulating regions that encoding conflict similarity, looking for the conjunction across several different measures of conflict encoding. These conflict measures use several best-practice approaches towards estimating representational similarity.*
- 1. The authors quantify several salient alternative hypothesis and systematically distinguish their core results from these alternatives.*
- 1. The question that the authors tackle is important to cognitive control, and they make a solid contribution.*

The authors have addressed several of my concerns. I appreciate the authors implementing best practices in their neuroimaging stats.

I think that the concerns that remain in my public review reflect the inherent limitations of the current work. The authors have done a good job working with the dataset they've collected.

Response: We would like to thank the reviewer for the positive evaluation of our manuscript and the constructive comments and suggestions. In response to your suggestions and concerns, we have removed the Stroop/Simon-only and the Stroop+Simon models, revised our conclusion and modified the misleading phrases.

We have provided detailed responses to your comments below.

- 1. The evidence from this previous work for mixtures between different conflict sources makes the framing of 'infinite possible types of conflict' feel like a strawman. The authors cite classic work (e.g., Kornblum et al., 1990) that develops a typology for conflict which is far from infinite. I think few people would argue that every possible source and level of difficulty will have to be learned separately. This work provides confirmatory evidence that task difficulty is represented parametrically (e.g., consistent with the n-back, MOT, and random dot motion literature).*

notes for my public concerns.

In their response, the authors say:

'If each combination of the Stroop-Simon combination is regarded as a conflict condition, there would be infinite combinations, and it is our major goal to investigate how these infinite conflict conditions are represented effectively in a space with finite dimensions.'

*I do think that this is a strawman. The paper doesn't make a strong case that this position ('infinite combinations') is widely held in the field. There is previous work (e.g., *n*-back, multiple object tracking, MSIT, dot motion) that has already shown parametric encoding of task difficulty. This paper provides confirmatory evidence, using an interesting new task, that demand are parametric, but does not provide a major theoretical advance.*

Response: We agree that the previous expression may have seemed somewhat exaggerative. While it is not “infinite”, recent research indeed suggests that the cognitive control shows domain-specificity across various “domains”, including conflict types (Egner, 2008), sensory modalities (Yang et al., 2017), task-irrelevant stimuli (Spape et al., 2008), and task sets (Hazeltine et al., 2011), to name a few.

These findings collectively support the notion that cognitive control is contextspecific (Bream et al., 2014). That is, cognitive control can be tuned and associated with different (and potentially large numbers of) contexts. Recently, Kikumoto and Mayr (2020) demonstrated that combinations of stimulus, rule and response in the same task formed separable, conjunctive representations. They further showed that these conjunctive representations facilitate performance. This is in line with the idea that each stimulus-location combination in the present task may be represented separately in a domain-specific manner. Moreover, domain-general task representation can also become domain-specific with learning, which further increases the number of domain-specific conjunctive representations (Mill et al., 2023). In line with the domain-specific account of cognitive control, we referred to the “infinite combinations” in our previous response to emphasize the extreme case of domainspecificity. However, recognizing that the term “infinite” may lead to ambiguity, we have replaced it with phrases such as “a large number of”, “highly varied”, in our revised manuscript.

We appreciate the reviewer for highlighting the potential connection of our work to existing literature that showed the parametric encoding of task difficulty (e.g., Dagher et al., 1999; Ritz & Shenhav, 2023). For instance, in Ritz et al.'s (2023) study, they parametrically manipulated target difficulty based on consistent ratios of dot color, and found that the difficulty was encoded in the caudal part of dorsal anterior cingulate cortex. Analogically, in our study, the “difficulty” pertains to the behavioral congruency effect that we modulated within the spatial Stroop and Simon dimensions. Notably, we did identify univariate effects in the right dmPFC and IPS associated with the difficulty in the Simon dimension. This parametric effect may lend support to our cognitive space hypothesis, although we exercised caution in interpreting their significance due to the absence of a clear brain-behavioral relevance in these regions. We have added the connection of our work to prior literature in the discussion. The parametric encoding of conflict also mirrors prior research showing the parametric encoding of task demands (Dagher et al., 1999; Ritz & Shenhav, 2023).

However, our analyses extend beyond solely testing the parametric encoding of difficulty. Instead, we focused on the multivariate representation of different conflict types, which we believe is independent from the univariate parametric encoding. Unlike the univariate encoding that relies on the strength within one dimension, the multivariate representation of conflict types incorporates both the spatial Stroop and Simon dimensions. Furthermore, we found that similar difficulty levels did not yield similar conflict representation, as indicated by the low similarity between the spatial Stroop and Simon conditions, despite both showing a similar level of congruency effect (Fig. S1). Additionally, we also observed an interaction

between conflict similarity and difficulty (i.e., congruency, Fig. 4B/D), such that the conflict similarity effect was more pronounced when conflict was present. Therefore, we believe that our findings make contribution to the literature beyond the difficulty effect.

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1. (Public Reviews) *The degree of Stroop vs Simon conflict is perfectly negatively correlated across conditions. This limits their interpretation of an integrated cognitive space, as they cannot separately measure Stroop and Simon effects. The author's control analyses have limited ability to overcome this task limitation. While these results are consistent with parametric encoding, they cannot adjudicate between combined vs separated representations.*

(Recommendations) *I think that it is still an issue that the task's two features (stroop and simon conflict) are perfectly correlated. This fundamentally limits their ability to measure the similarity in these features. The authors provide several control analyses, but I think these are limited.*

Response: We need to acknowledge that the spatial Stroop and Simon components in the five conflict conditions were not “perfectly” correlated, with $r = -0.89$. This leaves some room for the preliminary model comparison to adjudicate between these models. However, it's essential to note that conclusions based on these results must be tempered. In line with the reviewer's observation, we agree that the high correlation between the two conflict sources

posed a potential limitation on our ability to independently investigate the contribution of spatial Stroop and Simon conflicts. Therefore, in addition to the limitation we have previously acknowledged, we have now further revised our conclusion and adjusted our expressions accordingly.

Specifically, we now regard the parametric encoding of cognitive control not as direct evidence of the cognitive space view but as preliminary evidence that led us to propose this hypothesis, which requires further testing. Notably, we have also modified the title from “Conflicts are represented in a cognitive space to reconcile domain-general and domain-specific cognitive control” to “Conflicts are parametrically encoded: initial evidence for a cognitive space view to reconcile the debate of domain-general and domain-specific cognitive control”. Also, we revised the conclusion as: In sum, we showed that the cognitive control can be parametrically encoded in the right dlPFC and guides cognitive control to adjust goal-directed behavior. This finding suggests that different cognitive control states may be encoded in an abstract cognitive space, which reconciles the long-standing debate between the domain-general and domain-specific views of cognitive control and provides a parsimonious and more broadly applicable framework for understanding how our brains efficiently and flexibly represents multiple task settings.

From Recommendations The authors perform control analyses that test stroop-only and simon-only models. However, these analyses use a totally different similarity metric, that's based on set intersection rather than geometry. This metric had limited justification or explanation, and it's not clear whether these models fit worse because of the similarity metric. Even here, Simon-only model fit better than Stroop+Simon model. The dimensionality analyses may reflect the 1d manipulation by the authors (i.e. perfectly corrected stroop and simon effects).

Response: The Jaccard measure is the most suitable method we can conceive of for assessing the similarity between two conflicts when establishing the Stroop-only and Simon-only models, achieved by projecting them onto the vertical or horizontal axes, respectively (Author response image 1A). This approach offers two advantages. First, the Jaccard similarity combines both similarity (as reflected by the numerator) and distance (reflected by the difference between denominator and numerator) without bias towards either. Second, the Jaccard similarity in our design is equivalent to the cosine similarity because the denominator in the cosine similarity is identical to the denominator in the Jaccard similarity (both are the radius of the circle, Author response image 1B).

Author response image 1.

Definition of Jaccard similarity. A) Two conflicts (1 and 2) are projected onto the spatial Stroop/Simon axis in the Stroop/Simon-only model, respectively. The Jaccard similarity for

Stroop-only and Simon-only model are  and  respectively. Letters a-d are the projected

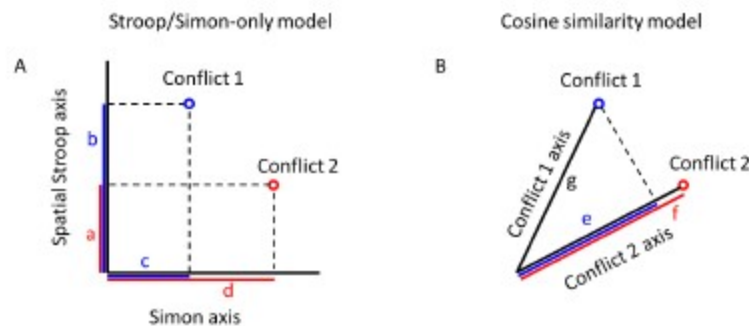
vectors from the two conflicts to the two axes. Blue and red colors indicate the conflict conditions. Shorter vectors are the intersection and longer vectors are the union. B)

According to the cosine similarity model, the similarity is defined as , where e is the

projected vector from conflict 1 to conflict 2, and g is the vector of conflict 1. The Jaccard

similarity for this case is defined by , where f is the projector vector from conflict 2 to itself.

Because $f = g$ in our design, the Jaccard similarity is equivalent to the cosine similarity.



Therefore, we believe that the model comparisons between cosine similarity model and the Stroop/Simon-Only models were equitable. However, we acknowledge the reviewer's and other reviewers' concerns about the correlation between spatial Stroop and Simon conflicts, which reduces the space to one dimension (1d) and limits our ability to distinguish between the Stroop-only and Simon-only models, as well as between Stroop+Simon and cosine similarity models. While these distinctions are undoubtedly important for understanding the geometry of the cognitive space, we recognize that they go beyond the major objective of this study, that is, to differentiate the cosine similarity model from domain-general/specific models. Therefore, we have chosen to exclude the Stroop-only, Simon-only and Stroop+Simon models in our revised manuscript.

Something that raised additional concerns are the RSMs in the key region of interest (Fig S5). The pure stroop task appears to be represented very differently from all of the conditions that include simon conflict.

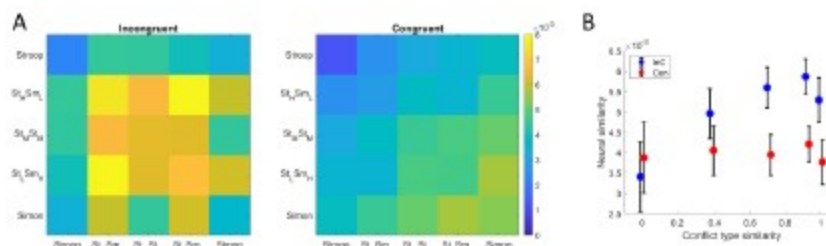
Together, I think these limitations reflect the structure of the task and research goals, not the statistical approach (which has been meaningfully improved).

Response: We appreciate the reviewer for pointing this out. It is essential to clarify that our conclusions were based on the significant similarity modulation effect identified in our statistical analysis using the cosine similarity model, where we did not distinguish between the within-Stroop condition and the other four within-conflict conditions (Fig. 7A, now Fig. 8A). This means that the representation of conflict type was not biased by the seemingly disparities in the values shown here. Moreover, to specifically test the differences between the within-Stroop condition and the other within-conflict conditions, we conducted a mixed-effect model analysis only including trial pairs from the same conflict type. In this analysis, the primary predictor was the cross-condition difference (0 for within-Stroop condition and 1 for other within-conflict conditions). The results showed no significant cross-condition difference in either the incongruent ($t = 1.22$, $p = .23$) or the congruent ($t = 1.06$, $p = .29$) trials. Thus, we believe the evidence for different similarities is inconclusive in our data and decided not to interpret this numerical difference. We have added this note in the revised figure caption for Figure S5.

Author response image 2.

Fig. S5. The stronger conflict type similarity effect in incongruent versus congruent conditions. (A) Summary representational similarity matrices for the right 8C region in incongruent (left) and congruent (right) conditions, respectively. Each cell represents the averaged Pearson correlation of cells with the same conflict type and congruency in the 1400×1400 matrix. Note that the seemingly disparities in the values of Stroop and other within-conflict cells (i.e., the diagonal) did not reach significance for either incongruent ($t = 1.22$, $p = .23$) or congruent ($t = 1.06$, $p = .29$) trials. (2) Scatter plot showing the averaged neural similarity (Pearson correlation) as a function of conflict type similarity in both conditions.

The values in both A and B are calculated from raw Pearson correlation values, in contrast to the z-scored values in Fig. 4D.



Minor:

- In the analysis of similarity_orientation, the df is very large (~14000). Here, and throughout, the df should be reflective of the population of subjects (ie be less than the sample size).

Response: The large degrees of freedom (df) in our analysis stem from the fact that we utilized a mixed-effect linear model, incorporating all data points (a total of $400 \times 35 = 14000$). In mixed-effect models, the df is determined by subtracting the number of fixed effects (in our case, 7) from the total number of observations. Notably, we are in line with the literature that have reported the df in this manner (e.g., Iravani et al., 2021; Schmidt & Weissman, 2015; Natraj et al., 2022).

Reference:

Iravani B, Schaefer M, Wilson DA, Arshamian A, Lundström JN. The human olfactory bulb processes odor valence representation and cues motor avoidance behavior. *Proc Natl Acad Sci U S A*. 2021 Oct 19;118(42):e2101209118. <https://doi.org/10.1073/pnas.2101209118>.

Schmidt, J.R., Weissman, D.H. Congruency sequence effects and previous response times: conflict adaptation or temporal learning?. *Psychological Research* 80, 590–607 (2016). <https://doi.org/10.1007/s00426-015-0681-x>.

Natraj, N., Silversmith, D. B., Chang, E. F., & Ganguly, K. (2022). Compartmentalized dynamics within a common multi-area mesoscale manifold represent a repertoire of human hand movements. *Neuron*, 110(1), 154-174. <https://doi.org/10.1016/j.neuron.2021.10.002>.

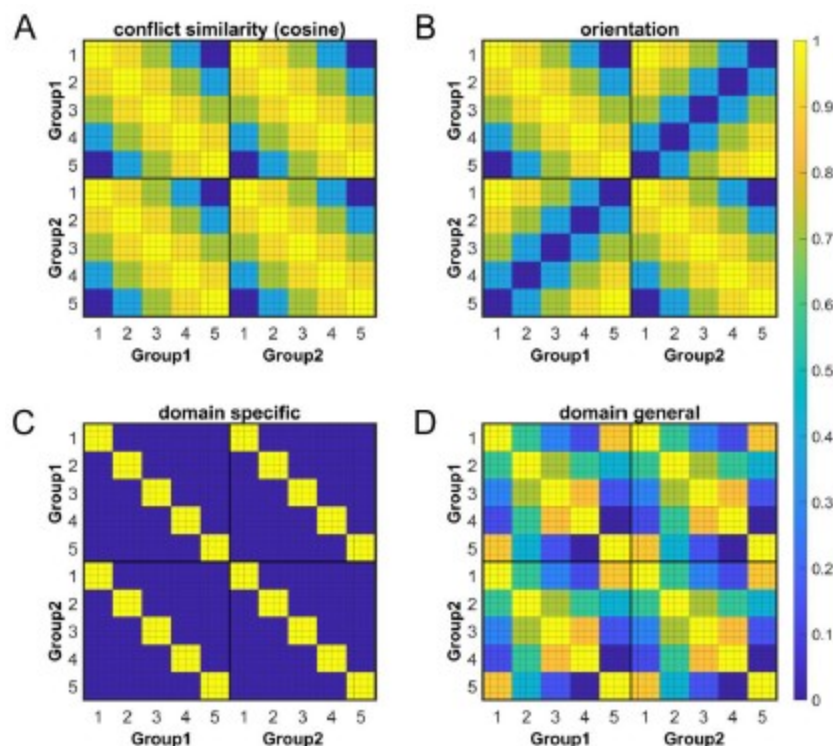
- it would improve the readability if there was more didactic justification for why analyses are done a certain way (eg justifying the jaccard metric). This will help less technically-savvy readers.

Response: We appreciate the reviewer's suggestion. However, considering the Stroop/Simon-only models in our design may not be a valid approach for distinguishing the contributions of the Stroop/Simon components, we have decided not to include the Jaccard metrics in our revised manuscript.

Besides, to improve the readability, we have moved Figure S4 to the main text (labeled as Figure 7), and added the domain-general/domain-specific schematics in Figure 8.

Author response image 3.

Figure 8. Schematic of key RSMs. (A) and (B) show the orthogonality between conflict similarity and orientation RSMs. The within-subject RSMs (e.g., Group1-Group1) for conflict similarity and orientation are all the same, but the cross-group correlations (e.g., Group2-Group1) are different. Therefore, we can separate the contribution of these two effects when including them as different regressors in the same linear regression model. (C) and (D) show the two alternative models. Like the cosine model (A), within-group trial pairs resemble between-group trial pairs in these two models. The domain-specific model is an identity matrix. The domain-general model is estimated from the absolute difference of behavioral congruency effect, but scaled to 0(lowest similarity)-1(highest similarity) to aid comparison. The plotted matrices here include only one subject each from Group 1 and Group 2. Numbers 1-5 indicate the conflict type conditions, for spatial Stroop, StHsmL, StMSmM, StLSmH, and Simon, respectively. The thin lines separate four different sub-conditions, i.e., target arrow (up, down) $\times$ congruency (incongruent, congruent), within each conflict type.



Reviewer #2:

This study examines the construct of "cognitive spaces" as they relate to neural coding schemes present in response conflict tasks. The authors use a novel experimental design in which different types of response conflict (spatial Stroop, Simon) are parametrically manipulated. These conflict types are hypothesized to be encoded jointly, within an abstract "cognitive space", in which distances between task conditions depend only on the similarity of conflict types (i.e., where conditions with similar relative proportions of spatial-Stroop versus Simon conflicts are represented with similar activity patterns). Authors contrast such a representational scheme for conflict with several other conceptually distinct schemes, including a domain-general, domain-specific, and two task-specific schemes. The authors conduct a behavioral and fMRI study to test which of these coding schemes is used by prefrontal cortex. Replicating the authors' prior work,

this study demonstrates that sequential behavioral adjustments (the congruency sequence effect) are modulated as a function of the similarity between conflict types. In fMRI data, univariate analyses identified activation in left prefrontal and dorsomedial frontal cortex that was modulated by the amount of Stroop or Simon conflict present, and representational similarity analyses (RSA) that identified coding of conflict similarity, as predicted under the cognitive space model, in right lateral prefrontal cortex.

This study tackles an important question regarding how distinct types of conflict might be encoded in the brain within a computationally efficient representational format. The ideas postulated by the authors are interesting ones and the statistical methods are generally rigorous.

Response: We would like to express our sincere appreciation for the reviewer's positive evaluation of our manuscript and the constructive comments and suggestions. In response to your suggestions and concerns, we excluded the StroopOnly, SimonOnly and Stroop+Simon models, and added the schematic of domain-general/specific model RSMs. We have provided detailed responses to your comments below.

The evidence supporting the authors claims, however, is limited by confounds in the experimental design and by lack of clarity in reporting the testing of alternative hypotheses within the method and results.

1. Model comparison

The authors commendably performed a model comparison within their study, in which they formalized alternative hypotheses to their cognitive space hypothesis. We greatly appreciate the motivation for this idea and think that it strengthened the manuscript. Nevertheless, some details of this model comparison were difficult for us to understand, which in turn has limited our understanding of the strength of the findings.

The text indicates the domain-general model was computed by taking the difference in congruency effects per conflict condition. Does this refer to the "absolute difference" between congruency effects? In the rest of this review, we assume that the absolute difference was indeed used, as using a signed difference would not make sense in this setting. Nevertheless, it may help readers to add this information to the text.

Response: We apologize for any confusion. The "difference" here indeed refers to the "absolute difference" between congruency effects. We have now clarified this by adding the word "absolute" accordingly.

"Therefore, we defined the domain-general matrix as the absolute difference in their congruency effects indexed by the group-averaged RT in Experiment 2."

Regarding the Stroop-Only and Simon-Only models, the motivation for using the Jaccard metric was unclear. From our reading, it seems that all of the other models --- the cognitive space model, the domain-general model, and the domain-specific model --- effectively use a Euclidean distance metric. (Although the cognitive space model is parameterized with cosine similarities, these similarity values are proportional to Euclidean distances because the points all lie on a circle. And, although the domain-general model is parameterized with absolute differences, the absolute difference is equivalent to Euclidean distance in 1D.) Given these considerations, the use of Jaccard seems to differ from the other models, in terms of parameterization, and thus potentially also in terms of underlying assumptions. Could authors help us understand why this distance metric was used instead of Euclidean distance? Additionally, if Jaccard must be

used because this metric seems to be non-standard in the use of RSA, it would likely be helpful for many readers to give a little more explanation about how it was calculated.

Response: We believe that the Jaccard similarity measure is consistent with the Cosine similarity measure. The Jaccard similarity is calculated as the intersection divided by the union. To define the similarity of two conflicts in the Stroop-only and Simon-only models, we first project them onto the vertical or horizontal axes, respectively (as shown in Author response image 1A). The Jaccard similarity in our design is equivalent to the cosine similarity because the denominator in the Jaccard similarity is identical to the denominator in the cosine similarity (both are the radius of the circle, Author response image 1B).

However, it is important to note that a cosine similarity cannot be defined when conflicts are projected onto spatial Stroop or Simon axis simultaneously. Therefore, we used the Jaccard similarity in the previous version of our manuscript.

Author response image 4.

Definition of Jaccard similarity. A) Two conflicts (1 and 2) are projected onto the spatial Stroop/Simon axis in the Stroop/Simon-only model, respectively. The Jaccard similarity for

Stroop-only and Simon-only model are $\frac{a}{b}$ and $\frac{c}{d}$ respectively. Letters a-d are the projected

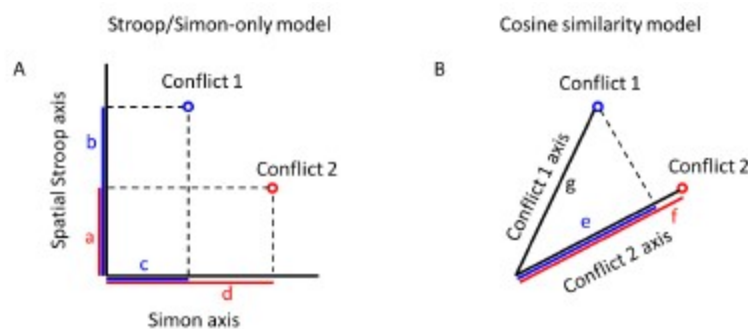
vectors from the two conflicts to the two axes. Blue and red colors indicate the conflict conditions. Shorter vectors are the intersection and longer vectors are the union. B)

According to the cosine similarity model, the similarity is defined as $\frac{e}{g}$, where e is the

projected vector from conflict 1 to conflict 2, and g is the vector of conflict 1. The Jaccard

similarity for this case is defined by $\frac{e}{f}$, where f is the projector vector from conflict 2 to itself.

Because $f = g$ in our design, the Jaccard similarity is equivalent to the cosine similarity.



However, we agree with the reviewer's and other reviewers' concern that the correlation between spatial Stroop and Simon conflicts makes it less likely to distinguish the Stroop+Simon from cosine similarity models. While distinguishing them is essential to understand the detailed geometry of the cognitive space, it is beyond our major purpose, that is, to distinguish the cosine similarity model with the domain-general/specific models. Therefore, we have chosen to exclude the Stroop-only, Simon-only and Stroop+Simon models from our revised manuscript.

When considering parameterizing the Stroop-Only and Simon-Only models with Euclidean distances, one concern we had is that the joint inclusion of these models might

render the cognitive space model unidentifiable due to collinearity (i.e., the sum of the Stroop-Only and Simon-Only models could be collinear with the cognitive space model). Could the authors determine whether this is the case? This issue seems to be important, as the presence of such collinearity would suggest to us that the design is incapable of discriminating those hypotheses as parameterized.

Response: We acknowledge that our design does not allow for a complete differentiation between the parallel encoding (StroopOnly+SimonOnly) model and the cognitive space model, given their high correlation ($r = 0.85$). However, it is important to note that the StroopOnly+SimonOnly model introduces more free parameters, making the model fitting poorer than the cognitive space model.

Additionally, the cognitive space model also shows high correlations with the StroopOnly and SimonOnly models (both $r_s = 0.66$). It is crucial to emphasize that our study's primary goal does not involve testing the parallel encoding hypothesis (through the StroopOnly+SimonOnly model). As a result, we have chosen to remove the model comparison results with the StroopOnly, SimonOnly and StroopOnly+SimonOnly models. Instead, the cognitive space model shows lower correlation with the purely domain-general ($r = -0.16$) and domain-specific ($r = 0.46$) models.

1. Issue of uniquely identifying conflict coding

We certainly appreciate the efforts that authors have taken to address potential confounders for encoding of conflict in their original submission. We broach this question not because we wish authors to conduct additional control analyses, but because this issue seems to be central to the thesis of the manuscript and we would value reading the authors' thoughts on this issue in the discussion.

To summarize our concerns, conflict seems to be a difficult variable to isolate within aggregate neural activity, at least relative to other variables typically studied in cognitive control, such as task-set or rule coding. This is because it seems reasonable to expect that many more nuisance factors covary with conflict -- such as univariate activation, level of cortical recruitment, performance measures, arousal --- than in comparison with, for example, a well-designed rule manipulation. Controlling for some of these factors post-hoc through regression is commendable (as authors have done here), but such a method will likely be incomplete and can provide no guarantees on the false positive rate.

Relatedly, the neural correlates of conflict coding in fMRI and other aggregate measures of neural activity are likely of heterogeneous provenance, potentially including rate coding (Fu et al., 2022), temporal coding (Smith et al., 2019), modulation of coding of other more concrete variables (Ebitz et al., 2020, 10.1101/2020.03.14.991745; see also discussion and reviews of Tang et al., 2016, 10.7554/eLife.12352), or neuromodulatory effects (e.g., Aston-Jones & Cohen, 2005). Some of these origins would seem to be consistent with "explicit" coding of conflict (conflict as a representation), but others would seem to be more consistent with epiphenomenal coding of conflict (i.e., conflict as an emergent process). Again, these concerns could apply to many variables as measured via fMRI, but at the same time, they seem to be more pernicious in the case of conflict. So, if authors consider these issues to be germane, perhaps they could explicitly state in the discussion whether adopting their cognitive space perspective implies a particular stance on these issues, how they interpret their results with respect to these issues, and if relevant, qualify their conclusions with uncertainty on these issues.

Response: We appreciate the reviewer's insightful comments regarding the representation and process of conflict.

First, we agree that the conflict is not simply a pure feature like a stimulus but often arises from the interaction (e.g., dimension overlap) between two or more aspects. For example, in the manual Stroop, conflict emerges from the inconsistent semantic information between color naming and word reading. Similarly, other higher-order cognitive processes such as task-set also underlie the relationship between concrete aspects. For instance, in a face/house categorization task, the taskset is the association between face/house and the responses. When studying these higher-order processes, it is often impossible to completely isolate them from bottomup features. Therefore, methods like the representational similarity analysis and regression models are among the limited tools available to attempt to dissociate these concrete factors from conflict representation. While not perfect, this approach has been suggested and utilized in practice (Freund et al., 2021).

Second, we agree that conflict can be both a representation and an emerging process. These two perspectives are not necessarily contradictory. According to David Marr's influential three-level theory (Marr, 1982), representation is the algorithm of the process to achieve a goal based on the input. Therefore, a representation can refer to not only a static stimulus (e.g., the visual representation of an image), but also a dynamic process. Building on this perspective, we posit that the representation of cognitive control consists of an array of dynamic representations embedded within the overall process. A similar idea has been proposed that the abstract task profiles can be progressively constructed as a representation in our brain (Kikumoto & Mayr, 2020).

We have incorporated this discussion into the manuscript:

"Recently an interesting debate has arisen concerning whether cognitive control should be considered as a process or a representation (Freund, Etzel, et al., 2021). Traditionally, cognitive control has been predominantly viewed as a process. However, the study of its representation has gained more and more attention. While it may not be as straightforward as the visual representation (e.g., creating a mental image from a real image in the visual area), cognitive control can have its own form of representation. An influential theory, Marr's (1982) three-level model proposed that representation serves as the algorithm of the process to achieve a goal based on the input. In other words, representation can encompass a dynamic process rather than being limited to static stimuli. Building on this perspective, we posit that the representation of cognitive control consists of an array of dynamic representations embedded within the overall process. A similar idea has been proposed that the representation of task profiles can be progressively constructed with time in the brain (Kikumoto & Mayr, 2020)."

Reference:

Freund, M. C., Etzel, J. A., & Braver, T. S. (2021). Neural Coding of Cognitive Control: The Representational Similarity Analysis Approach. *Trends in Cognitive Sciences*, 25(7), 622-638. <https://doi.org/10.1016/j.tics.2021.03.011>

Marr, D. C. (1982). *Vision: A computational investigation into human representation and information processing*. New York: W.H. Freeman.

Kikumoto A, Mayr U. (2020). Conjunctive representations that integrate stimuli, responses, and rules are critical for action selection. *Proceedings of the National Academy of Sciences*, 117(19):10603-10608. <https://doi.org/10.1073/pnas.1922166117>.

1. Interpretation of measured geometry in 8C

We appreciate the inclusion of the measured similarity matrices of area 8C, the key area the results focus on, to the supplemental, as this allows for a relatively model-agnostic

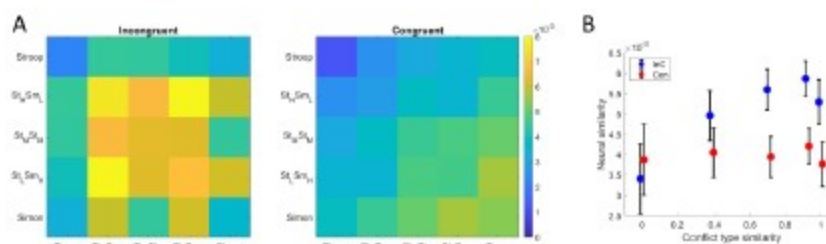
look at a portion of the data. Interestingly, the measured similarity matrix seems to mismatch the cognitive space model in a potentially substantive way. Although the model predicts that the "pure" Stroop and Simon conditions will have maximal self-similarity (i.e., the Stroop-Stroop and Simon-Simon cells on the diagonal), these correlations actually seem to be the lowest, by what appears to be a substantial margin (particularly the Stroop-Stroop similarities). What should readers make of this apparent mismatch? Perhaps authors could offer their interpretation on how this mismatch could fit with their conclusions.

Response: We appreciate the reviewer for bringing this to our attention. It is essential to clarify that our conclusions were based on the significant similarity modulation effect observed in our statistical analysis using the cosine similarity model, where we did not distinguish between the within-Stroop condition and the other four within-conflict conditions (Fig. 7A). This means that the representation of conflict type was not biased by the seemingly disparities in the values shown here. Moreover, to specifically address the potential differences between the within-Stroop condition and the other within-conflict conditions, we conducted a mixed-effect model. In this analysis, the primary predictor was the cross-condition difference (0 for within-Stroop condition and 1 for other within-conflict conditions). The results showed no significant cross-condition difference in either the incongruent trials ($t = 1.22$, $p = .23$) or the congruent ($t = 1.06$, $p = .29$) trials. Thus, we believe the evidence for different similarities is inconclusive in our data and decided not to interpret this numerical difference.

We have added this note in the revised figure caption for Figure S5.

Author response image 5.

Fig. S5. The stronger conflict type similarity effect in incongruent versus congruent conditions. (A) Summary representational similarity matrices for the right 8C region in incongruent (left) and congruent (right) conditions, respectively. Each cell represents the averaged Pearson correlation of cells with the same conflict type and congruency in the 1400×1400 matrix. Note that the seemingly disparities in the values of Stroop and other within-conflict cells (i.e., the diagonal) did not reach significance for either incongruent ($t = 1.22$, $p = .23$) or congruent ($t = 1.06$, $p = .29$) trials. (2) Scatter plot showing the averaged neural similarity (Pearson correlation) as a function of conflict type similarity in both conditions. The values in both A and B are calculated from raw Pearson correlation values, in contrast to the z-scored values in Fig. 4D.

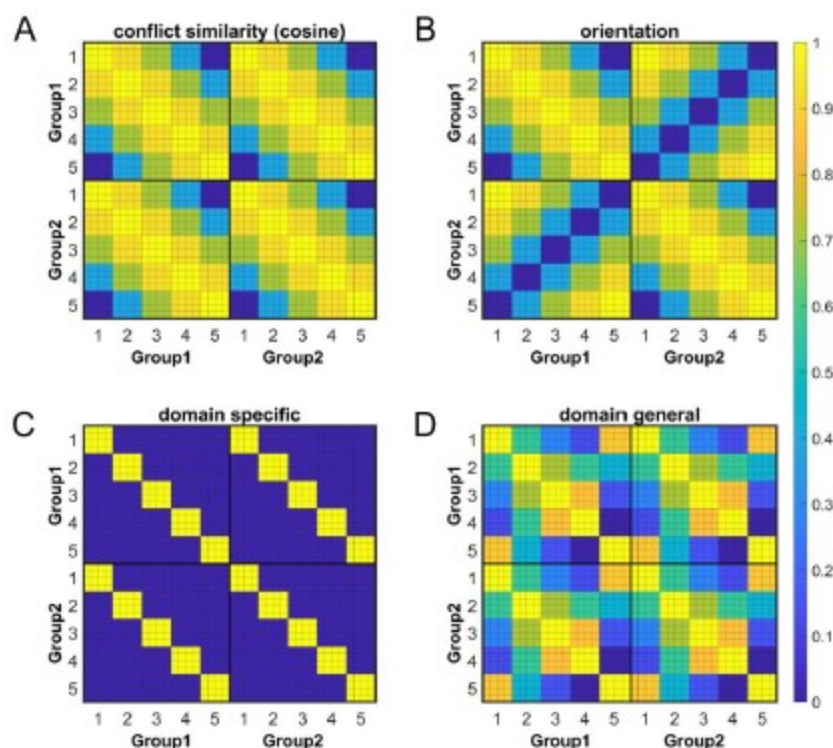


1. It would likely improve clarity if all of the competing models were displayed as summarized RSA matrices in a single figure, similar to (or perhaps combined with) Figure 7.

Response: We appreciate the reviewer's suggestion. We now have incorporated the domain-general and domain-specific models into the Figure 7 (now Figure 8).

Author response image 6.

Figure 8. Schematic of key RSMs. (A) and (B) show the orthogonality between conflict similarity and orientation RSMs. The within-subject RSMs (e.g., Group1-Group1) for conflict similarity and orientation are all the same, but the cross-group correlations (e.g., Group2-Group1) are different. Therefore, we can separate the contribution of these two effects when including them as different regressors in the same linear regression model. (C) and (D) show the two alternative models. Like the cosine model (A), within-group trial pairs resemble between-group trial pairs in these two models. The domain-specific model is an identity matrix. The domain-general model is estimated from the absolute difference of behavioral congruency effect, but scaled to 0(lowest similarity)-1(highest similarity) to aid comparison. The plotted matrices here include only one subject each from Group 1 and Group 2. Numbers 1-5 indicate the conflict type conditions, for spatial Stroop, StHsmL, StMSmM, StLSmH, and Simon, respectively. The thin lines separate four different sub-conditions, i.e., target arrow (up, down) $\times$ congruency (incongruent, congruent), within each conflict type.



1. Because this model comparison is key to the main inferences in the study, it might also be helpful for most readers to move all of these RSA model matrices to the main text, instead of in the supplemental.

Response: We thank the reviewer for this suggestion. We have moved the Fig. S4 to the main text, labeled as the new Figure 7.

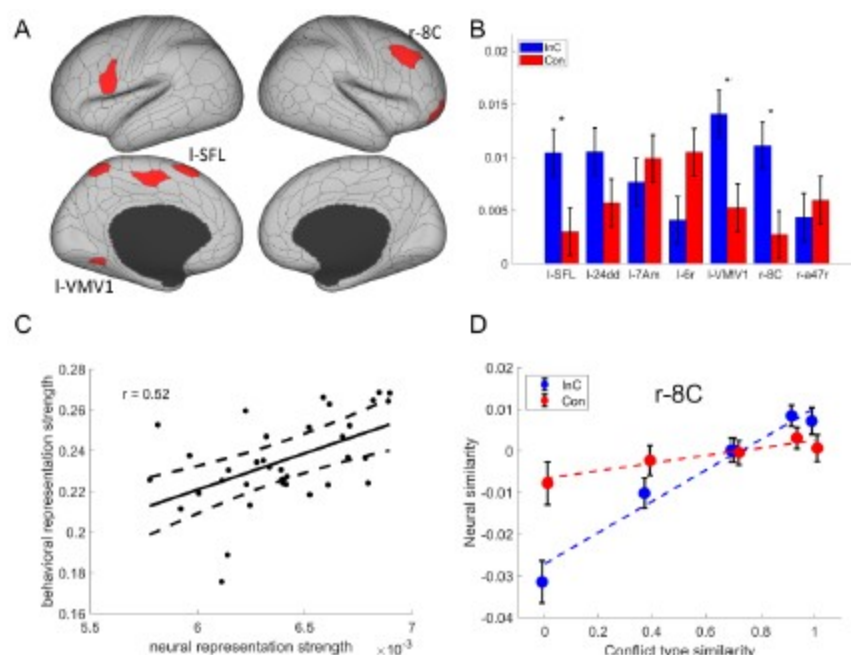
1. It may be worthwhile to check how robust the observed brain-behavior association (Fig 4C) is to the exclusion of the two datapoints with the lowest neural representation strength measure, as these points look like they have high leverage.

Response: We calculated the Pearson correlation after excluding the two points and found it does not affect the results too much, with the $r = 0.50$, $p = .003$ (compared to the original $r = 0.52$, $p = .001$).

Additionally, we found the two axes were mistakenly shifted in Fig 4C. Therefore, we corrected this error in the revised manuscript. The correlation results would not be influenced.

Author response image 7.

Fig. 4. The conflict type effect. (A) Brain regions surviving the Bonferroni correction ($p < 0.0001$) across the regions (criterion 1). Labeled regions are those meeting the criterion 2. (B) Different encoding of conflict type in the incongruent with congruent conditions. * Bonferroni corrected $p < .05$. (C) The brain-behavior correlation of the right 8C (criterion 3). The x-axis shows the beta coefficient of the conflict type effect from the RSA, and the y-axis shows the beta coefficient obtained from the behavioral linear model using the conflict similarity to predict the CSE in Experiment 2. (D) Illustration of the different encoding strength of conflict type similarity in incongruent versus congruent conditions of right 8C. The y-axis is derived from the z-scored Pearson correlation coefficient, consistent with the RSA methodology. See Fig. S4B for a plot with the raw Pearson correlation measurement. l = left; r = right.



Reviewer #3:

Yang and colleagues investigated whether information on two task-irrelevant features that induce response conflict is represented in a common cognitive space. To test this, the authors used a task that combines the spatial Stroop conflict and the Simon effect. This task reliably produces a beautiful graded congruency sequence effect (CSE), where the cost of congruency is reduced after incongruent trials. The authors measured fMRI to identify brain regions that represent the graded similarity of conflict types, the congruency of responses, and the visual features that induce conflicts. They applied univariate, multivariate, and connectivity analyses to fMRI data to identify brain regions that represent the graded similarity of conflict types, the congruency of responses, and

the visual features that induce conflicts. They further directly assessed the dimensionality of represented conflict space.

The authors identified the right dlPFC (right 8C), which shows 1) stronger encoding of graded similarity of conflicts in incongruent trials and 2) a positive correlation between the strength of conflict similarity type and the CSE on behavior. The dlPFC has been shown to be important for cognitive control tasks. As the dlPFC did not show a univariate parametric modulation based on the higher or lower component of one type of conflict (e.g., having more spatial Stroop conflict or less Simon conflict), it implies that dissimilarity of conflicts is represented by a linear increase or decrease of neural responses. Therefore, the similarity of conflict is represented in multivariate neural responses that combine two sources of conflict.

The strength of the current approach lies in the clear effect of parametric modulation of conflict similarity across different conflict types. The authors employed a clever cross-subject RSA that counterbalanced and isolated the targeted effect of conflict similarity, decorrelating orientation similarity of stimulus positions that would otherwise be correlated with conflict similarity. A pattern of neural response seems to exist that maps different types of conflict, where each type is defined by the parametric gradation of the yoked spatial Stroop conflict and the Simon conflict on a similarity scale. The similarity of patterns increases in incongruent trials and is correlated with CSE modulation of behavior.

The main significance of the paper lies in the evidence supporting the use of an organized "cognitive space" to represent conflict information as a general control strategy. The authors thoroughly test this idea using multiple approaches and provide convincing support for their findings. However, the universality of this cognitive strategy remains an open question.

(Public Reviews) Taken together, this study presents an exciting possibility that information requiring high levels of cognitive control could be flexibly mapped into cognitive map-like representations that both benefit and bias our behavior. Further characterization of the representational geometry and generalization of the current results look promising ways to understand representations for cognitive control.

Response: We would like to thank the reviewer for the positive evaluation of our manuscript and for providing constructive comments. In response to your suggestions, we have acknowledged the potential limitation of the design and the cross-subject RSA approach, and incorporated the open questions to the discussions. Please find our detailed responses below.

The task presented in the study involved two sources of conflict information through a single salient visual input, which might have encouraged the utilization of a common space.

Response: We agree that the unified visual input in our design may have facilitated the utilization of a common space. However, we believe the stimuli are not necessarily unified in the construction of the common space. To further test the potential interaction between the concrete stimulus setting and the cognitive space representation, it is necessary to use varied stimuli in future research. We have left this as an open question in the discussion:

Can we effectively map any sources of conflict with completely different stimuli into a single space?

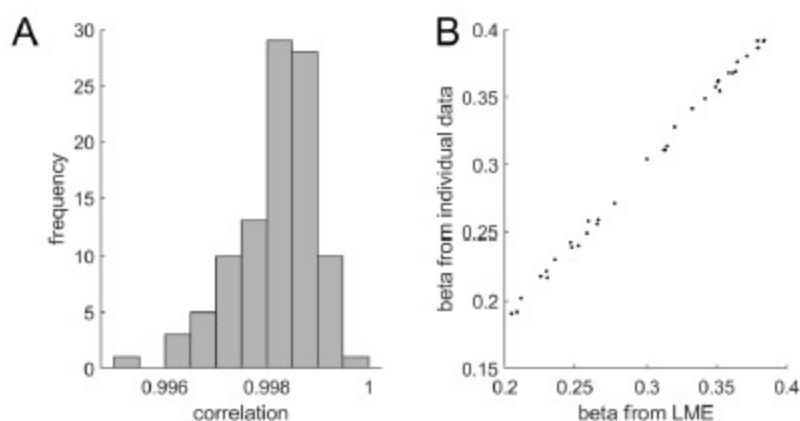
The similarity space was analyzed at the level of between-individuals (i.e., crosssubject RSA) to mitigate potential confounds in the design, such as congruency and the orientation of stimulus positions. This approach makes it challenging to establish a direct

link between the quality of conflict space representation and the patterns of behavioral adaptations within individuals.

Response: By setting the variables as random effects at the subject level, we have extracted the individual effects that incorporate both the group-level fixed effects and individual-level random effects. We believe this approach yields results that are as reliable, if not more, than effects calculated from individual data only. First, the mixed effect linear (LME) model has included all the individual data, forming the basis for establishing random effects. Therefore, the individual effects derived from this approach inherently reflect the individual-specific effects. To support this notion, we have included a simulation script (accessible in the online file “simulation_LME.mlx” at <https://osf.io/rcq8w>) to demonstrate the strong consistency between the two approaches (see Author response image 8). In this simulation, we generated random data (Y) for 35 subjects, each containing 20 repeated measurements across 5 conditions. To streamline the simulation, we only included one predictor (X), which was treated as both fixed and random effects at the subject level. We applied two methods to calculate the individual beta coefficient. The first involved extracting individual beta coefficients from the LME model by summing the fixed effect with the subject-specific random effect. The second method was entailed conducting a regression analysis using data from each subject to obtain the slope. We tested their consistency by calculating the Pearson correlation between the derived beta coefficients. This simulation was repeated 100 times.

Author response image 8.

The consistent individual beta coefficients between the mixed effect model and the individual regression analysis. A) The distribution of Pearson correlation between the two methods for 100 times. B) An example from the simulation showing the highly correlated results from the two methods. Each data point indicates a subject (n=35).



Second, the potential difference between the two methods lies in that the LME model have also taken the group-level variance into account, such as the dissociable variances of the conflict similarity and orientation across subject groups. This enabled us to extract relatively cleaner conflict similarity effects for each subject, which we believe can be better linked to the individual behavioral adaptations. Moreover, we have extracted the behavioral adaptations scores (i.e., the similarity modulation effect on CSE) using a similar LME approach. Conducting behavioral analysis solely using individual data would have been less reliable, given the limited sample size of individual data (~32 points per subject). This also motivated us to maintain consistency by extracting individual neural effects using LME models.

Furthermore, it remains unclear at which cognitive stages during response selection such a unified space is recruited. Can we effectively map any sources of conflict into a single scale? Is this unified space adaptively adjusted within the same brain region? Additionally, does the amount of conflict solely define the dimensions of this unified space across many conflict-inducing tasks? These questions remain open for future studies to address.

Response: We appreciate the reviewer's constructive open questions. We respond to each of them based on our current understanding.

1. It remains unclear at which cognitive stages during response selection such a unified space is recruited.

We anticipate that the cognitive space is recruited to guide the transference of behavioral CSE at two critical stages. The first stage involves the evaluation of control demands, where the representational distance/similarity between previous and current trials influences the adjustment of cognitive control. The second stage pertains to is control execution, where the switch from one control state to another follows a path within the cognitive space. It is worth noting that future studies aiming to address this question may benefit from methodologies with higher temporal resolutions, such as EEG and MEG, to provide more precise insights into the temporal dynamics of the process of cognitive space recruitment.

1. Can we effectively map any sources of conflict into a single scale?

It is possible that various sources of conflict can be mapped onto the same space based on their similarity, even if finding such an operational defined similarity may be challenging. However, our results may offer an approach to infer the similarity between two conflicts. One way is to examine their congruency sequence effect (CSE), with a stronger CSE suggesting greater similarity. The other way is to test their representational similarity within the dorsolateral prefrontal cortex.

1. Is this unified space adaptively adjusted within the same brain region? We do not have an answer to this question. We showed that the cognitive space does not change with time (Note. S3). What have adjusted is the control demand to resolve the quickly changing conflict conditions from trial to trial. Though, it is an interesting question whether the cognitive space may be altered, for example, when the mental state changes significantly. And if yes, we can further test whether the change of cognitive space is also within the right dlPFC.
2. Additionally, does the amount of conflict solely define the dimensions of this unified space across many conflict-inducing tasks?

Our understanding of this comment is that the amount of conflict refers to the number of conflict sources. Based on our current finding, the dimensions of the space are indeed defined by how many different conflict sources are included. However, this would require the different conflict sources are orthogonal. If some sources share some aspects, the cognitive space may collapse to a lower dimension. We have incorporated the first question into the discussion:

Moreover, we anticipate that the representation of cognitive space is most prominently involved at two critical stages to guide the transference of behavioral CSE. The first stage involves the evaluation of control demands, where the representational distance/similarity between previous and current trials influences the adjustment of cognitive control. The

second stage pertains to control execution, where the switch from one control state to another follows a path within the cognitive space. However, we were unable to fully distinguish between these two stages due to the low temporal resolution of fMRI signals in our study. Future research seeking to delve deeper into this question may benefit from methodologies with higher temporal resolutions, such as EEG and MEG.

We have included the other questions into the manuscript as open questions, calling for future research.

Several interesting questions remains to be answered. For example, is the dimension of the unified space across conflict-inducing tasks solely determined by the number of conflict sources? Can we effectively map any sources of conflict with completely different stimuli into a single space? Is the cognitive space geometry modulated by the mental state? If yes, what brain regions mediate the change of cognitive space?

Minor comments:

- *The original comment about out-of-sample predictions to examine the continuity of the space was a suggestion for testing neural representations, not behavior (I apologize for the lack of clarity). Given the low dimensionality of the conflict space shown by the participation ratio, we expect that linear separability exists only among specific combinations of conditions. For example, the pair of conflicts 1 and 5 together is not linearly separable from conflicts 2 and 3. But combined with other results, this is already implied.*

Response: We apologize for the misunderstanding. In fact, performing a prediction analysis using the extensive RSM in our study does presents certain challenges, primarily due to its substantial size (1400x1400) and the intricate nature of the mixed-effect linear model. In our efforts to simplify the prediction process by excluding random effects, we did observe a correlation between the predicted and original values, albeit a relatively small Pearson correlation coefficient of $r = 0.024$, $p < .001$. This small correlation can be attributed to two key factors. First, the exclusion of data points impacts not only the conflict similarity regressor but also other regressors within the model, thereby diminishing the predictive power. Secondly, the large amount of data points in the model heightens the risk of overfitting, subsequently reducing the model's capacity for generalization and increasing the likelihood of unreliable predictions. Given these potential problems, we have opted not to include this prediction in the revised manuscript.